

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

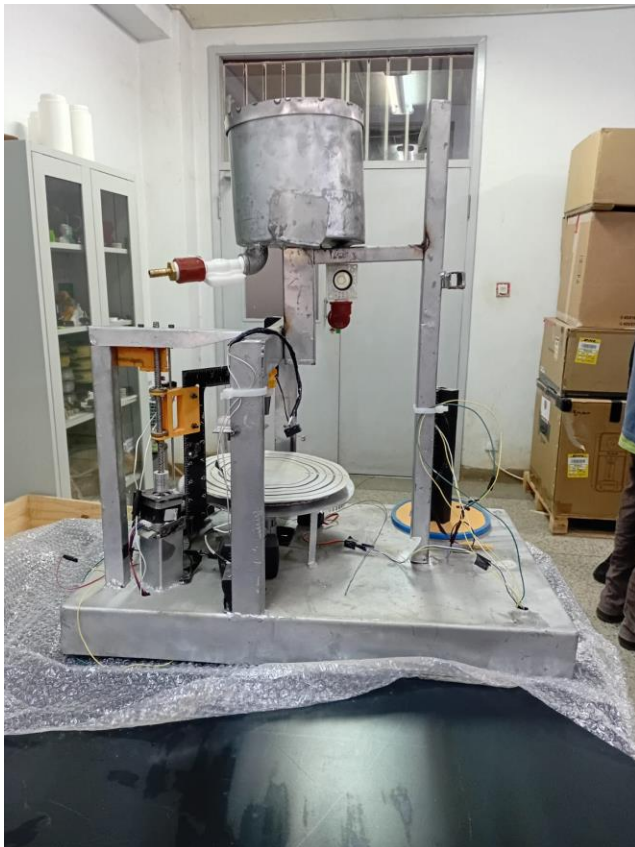
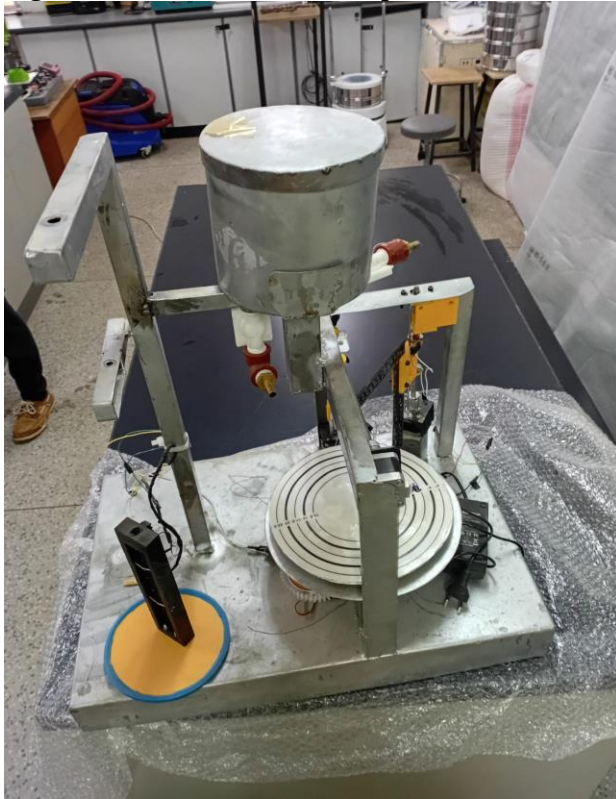
The application of frosting on cakes is an important finishing process within the commercial baking and confectionery industry. Frosting serves both functional and decorative purposes, including moisture retention, surface protection, structural layering, and visual enhancement of baked products. In commercial bakery operations, frosting quality is commonly evaluated based on parameters such as surface uniformity, frosting thickness consistency, edge smoothness, and repeatability of decorative patterns. Achieving these characteristics manually requires considerable operator skill and experience, particularly when handling high-viscosity frosting materials such as whipped cream.

Cake frosting materials exhibit non-Newtonian flow behavior characterized by yield stress and shear-thinning properties. Unlike Newtonian fluids, cake frosting does not flow uniformly under low applied stress and requires sufficient extrusion pressure before material displacement occurs. Variations in the frosting composition, preparation method, and ambient temperature may also influence viscosity and extrusion characteristics. These properties introduce engineering challenges in automated frosting systems, particularly in relation to pressure stability, extrusion lag, nozzle flow consistency, and repeatable volumetric deposition.

Recent developments in food automation and computer numerical control systems have enabled the application of robotic and programmable motion systems in culinary processes such as chocolate deposition, food printing, icing decoration, and precision dispensing operations. Several automated cake decoration systems have been proposed using Cartesian motion platforms, various extrusion systems, and rotational cake positioning mechanisms. However, many existing low-cost systems experience limitations related to structural rigidity, insufficient extrusion force, inconsistent frosting thickness, and poor synchronization between rotational and linear motion systems when handling highly viscous frosting materials.

The previous project conducted on the automated cake frosting process, utilized a whipped-cream-based extrusion approach combined with a simplified motion system for frosting application. Although the system demonstrated the feasibility of automated frosting deposition, several operational limitations were identified. These included limited extrusion capability for higher-viscosity frosting materials, inconsistent frosting application, and insufficient control of rotational synchronization during cylindrical surface coating. Furthermore, the previous system was not optimized for other types of frosting, whose rheological behavior requires significantly

higher extrusion force and improved volumetric flow regulation.



The current project focuses on the modification and improvement of the previous automated frosting system through the integration of a multi-axis Cartesian flying gantry mechanism, a Theta-axis rotary platform, and a positive displacement piston-based extrusion subsystem. The design seeks to improve frosting deposition accuracy, rotational coordination, structural rigidity, and extrusion stability for frosting applications. The system also incorporates synchronized multi-axis motion control using an ESP32-S3 controller to coordinate frosting application on both the top and side surfaces of cylindrical cakes.

By combining CNC-based motion control with controlled volumetric extrusion principles, the project contributes toward the development of automated culinary deposition systems capable of handling high-viscosity non-Newtonian food materials under repeatable operating conditions.

1.2 Problem Statement

The automated deposition of high-viscosity frosting materials onto cylindrical cake surfaces presents significant engineering challenges related to extrusion stability, synchronized motion control, and frosting thickness uniformity. Cake frosting exhibits non-Newtonian behavior characterized by yield stress and shear-thinning properties, requiring sufficient extrusion force and stable pressure regulation to achieve continuous and controlled material flow through a nozzle system.

Conventional manual frosting processes are highly dependent on operator skill and often produce inconsistencies in frosting thickness, surface smoothness, and edge uniformity. Existing low-cost automated frosting systems have also demonstrated limitations in handling high-viscosity frosting materials due to inadequate extrusion mechanisms, poor synchronization between rotational and linear motion systems, and insufficient structural rigidity during operation.

In the previous automated frosting project, the use of whipped-cream-oriented extrusion and simplified motion architecture limited the system's ability to accommodate other types of frosting and achieve uniform frosting application. Challenges related to rotational coordination, extrusion lag, and controlled volumetric deposition remained insufficiently addressed.

There is therefore a need for an improved automated frosting system capable of controlled volumetric deposition of frosting on various cake geometries through synchronized multi-axis motion control, stable extrusion pressure generation, and coordinated rotational frosting application.

1.3 Main Objective

To design and develop a multi-axis automated cake frosting system capable of controlled volumetric deposition of cake frosting on round cakes with diameters of up to 8 inches.

1.4 Specific Objectives

1. To improve the structural rigidity, rotational synchronization, and frosting application capability of the previous automated cake frosting system.
2. To design a multi-axis motion system capable of coordinating nozzle motion and cake positioning for frosting application on cylindrical cake geometries.
3. To design a frosting delivery system capable of providing controlled and consistent deposition of frosting materials during automated cake decoration.
4. To develop a control system capable of coordinating motion, extrusion, and rotational operations within the automated frosting machine.
5. To calibrate and integrate the different subsystems.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The automation of cake frosting processes combines principles from food engineering, computer numerical control (CNC) systems, fluid mechanics, motion control, and mechatronic system integration. Automated frosting systems must coordinate controlled material extrusion with synchronized multi-axis motion in order to achieve repeatable frosting thickness and surface uniformity. The complexity of these systems increases significantly when handling high-viscosity non-Newtonian frosting materials such as buttercream, whose rheological behavior introduces additional requirements related to extrusion pressure, nozzle geometry, and flow stability.

Several studies have explored the application of robotic systems and additive manufacturing techniques in food processing operations. Research in food printing, icing deposition, and automated cake decoration has demonstrated the potential of programmable motion systems in improving repeatability and reducing operator dependency. However, limitations related to extrusion force capability, rotational synchronization, structural rigidity, and frosting consistency remain significant challenges in low-cost automated frosting systems.

This chapter reviews existing literature related to automated cake decoration systems, frosting rheology, extrusion mechanisms, rotary kinematics and CNC motion architectures relevant to the development of the automated cake frosting machine.

2.2 Automated Cake Frosting Systems

Cake frosting is an essential stage in cake production that influences both the visual appearance and overall quality of the final product. The frosting process involves the application of a frosting material onto the surfaces of a cake to produce a smooth finish, decorative features, and a protective layer that enhances product presentation and shelf appeal. Traditionally, frosting operations have been performed manually by skilled decorators using spatulas, piping bags, and specialized decorating tools. While manual frosting remains common in many bakeries, increasing production demands have encouraged the adoption of automated frosting technologies.

2.2.1 Manual Cake Frosting

Manual frosting relies heavily on the skill, experience, and consistency of the operator. The decorator applies frosting to the top and side surfaces of a cake while simultaneously controlling frosting thickness, surface smoothness, and decorative quality. Although manual frosting offers flexibility and creativity, several limitations are associated with the process.

These limitations include:

- variation in frosting thickness between operators,
- inconsistent surface finish quality,
- relatively slow production rates,
- operator fatigue during extended production,
- difficulty maintaining repeatability across large production batches,
- use of manual rotary tables.

As production volumes increase, maintaining consistent frosting quality through manual methods becomes increasingly challenging. Consequently, automated frosting systems have been investigated as a means of improving consistency and reducing dependence on operator skill.



2.2.2 Semi-Automated Cake Frosting Systems

Semi-automated frosting systems combine mechanical assistance with operator intervention. In such systems, the cake may be rotated automatically while frosting is manually applied, or frosting delivery may be automated while the operator controls nozzle positioning.

Typical semi-automated systems incorporate:

- rotating electric cake turntables,
- frosting dispensers,
- manually guided frosting applicators,
- automated smoothing tools.

These systems improve production efficiency compared to fully manual frosting while retaining some level of operator control. However, frosting quality remains partially dependent on user skill, and repeatability may still be difficult to achieve for large-scale production.

Semi-automated systems are commonly used in small and medium-sized bakeries where production volumes do not justify the investment associated with fully automated equipment.



2.2.3 Fully Automated Cake Frosting Systems

Advances in automation, robotics, and computer numerical control (CNC) technologies have created opportunities for the development of fully automated cake frosting systems capable of performing frosting operations with minimal operator intervention. While traditional cake decoration remains heavily dependent on manual labour and operator skill, recent developments in automated food deposition systems have demonstrated the potential for robotic frosting and decoration processes.

The primary goals of automated frosting technology include:

- consistent frosting thickness,
- repeatable surface finishing,
- reduced labour requirements,
- improved production throughput,
- enhanced product uniformity,
- programmable decoration capability.

Various approaches have been proposed for automated frosting application. Some systems utilize moving nozzles that deposit frosting onto stationary cakes, while others employ rotating cake platforms combined with coordinated nozzle movement. More advanced concepts integrate multiple motion axes, allowing simultaneous control of nozzle position, cake rotation, and frosting delivery.

2.3 Frosting Materials and Rheological Behaviour

The performance of automated food deposition systems is strongly influenced by the physical and flow characteristics of the material being processed. In cake frosting applications, the deposited material must be capable of flowing through a delivery system while simultaneously retaining its shape after deposition. These requirements make frosting materials fundamentally different from simple liquids and introduce significant challenges in the design of automated extrusion systems.

The study of material deformation and flow is known as rheology. Rheology provides a framework for understanding how materials respond to applied forces and is widely used in food engineering to analyze products such as sauces, doughs, creams, icing, frosting, chocolate, and other semi-solid food materials. Knowledge of rheological behaviour is essential for predicting pressure requirements, extrusion performance, flow stability, and deposition quality within automated food processing systems.

2.3.1 Classification of Food Fluids

Food materials may generally be classified according to the relationship between the applied force and the resulting flow behaviour. Many liquid food products exhibit behaviour similar to conventional fluids and can be described using classical fluid mechanics principles. However, numerous food materials demonstrate more complex responses that cannot be adequately described using simple fluid models.

Food materials commonly encountered in processing operations include:

- liquids,
- suspensions,
- emulsions,
- gels,
- pastes,
- semi-solid materials.

Examples include:

Material Type	Example
Liquid	Water, milk
Suspension	Fruit juice with pulp
Emulsion	Mayonnaise
Gel	Jelly
Paste	Tomato paste
Semi-solid	Frosting, icing, dough

The flow characteristics of these materials vary significantly and influence the selection of pumping, dispensing, and extrusion systems.

2.3.2 Newtonian Fluids

Newtonian fluids are materials whose viscosity remains constant regardless of the rate at which they are deformed. In these fluids, the relationship between shear stress and shear rate is linear, meaning that increasing the applied force results in a proportional increase in flow.

Examples of Newtonian fluids include:

- water,
- vegetable oils,
- glycerine,
- and many low-viscosity liquids.

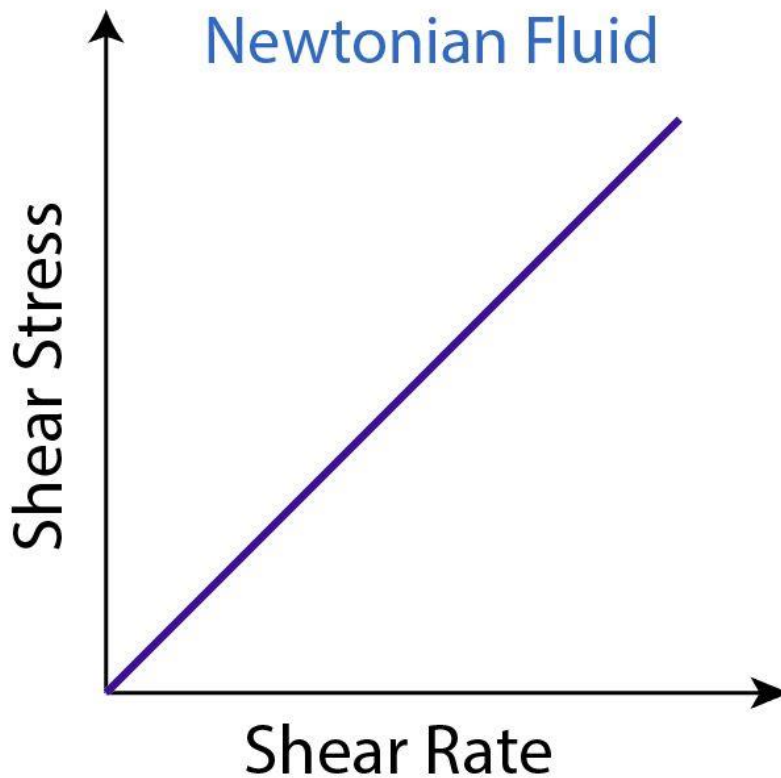
The relationship between shear stress and shear rate for a Newtonian fluid is expressed as:

$$\tau = \mu \dot{\gamma}$$

where:

- τ is the shear stress,
- μ is the dynamic viscosity,
- $\dot{\gamma}$ is the shear rate.

For Newtonian fluids, viscosity remains constant and is independent of the applied shear rate.



The predictable behaviour of Newtonian fluids simplifies flow analysis and equipment design. However, many food materials do not exhibit this behaviour.

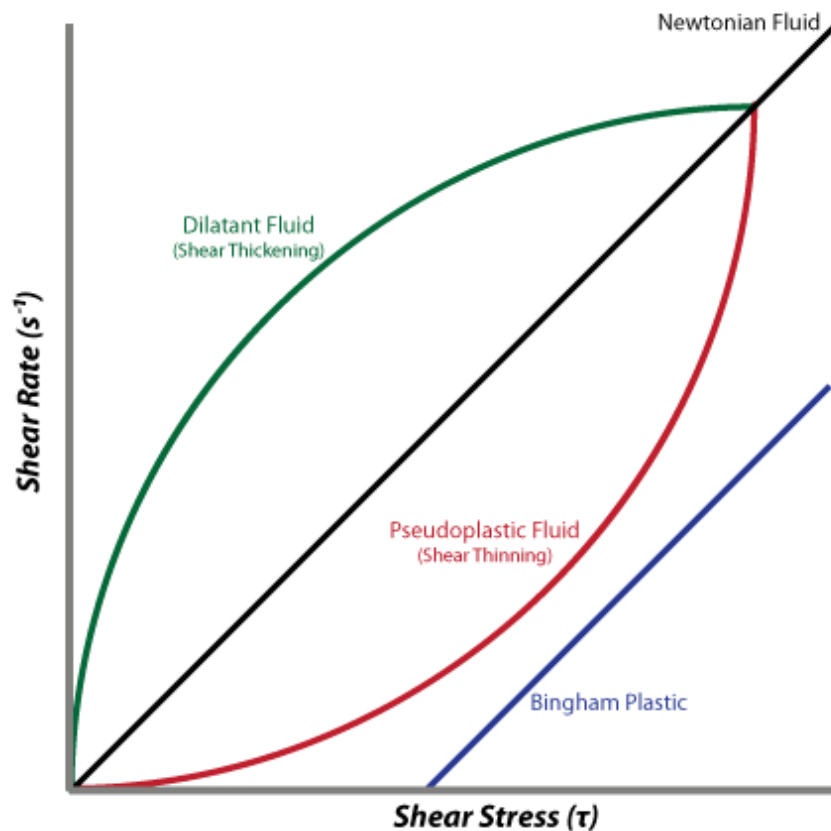
2.3.3 Non-Newtonian Fluids

Non-Newtonian fluids exhibit viscosities that change when subjected to varying levels of shear. Unlike Newtonian fluids, the relationship between shear stress and shear rate is nonlinear and may depend on factors such as deformation history, temperature, composition, and processing conditions.

Many food products exhibit non-Newtonian behaviour, including:

- ketchup,
- yoghurt,
- mayonnaise,
- chocolate,
- dough,
- frosting materials,
- icing formulations.

Because viscosity changes during flow, non-Newtonian materials often require specialized pumping and extrusion systems.



2.3.4 Shear-Thinning Behaviour

Shear-thinning behaviour occurs when a material's apparent viscosity decreases as the applied shear rate increases. In practical terms, the material becomes easier to flow when subjected to higher deformation rates.

Many food products demonstrate shear-thinning characteristics because their internal structures become progressively aligned with the direction of flow under increasing shear.

Examples include:

- ketchup,
- yoghurt,
- mayonnaise,
- toothpaste,
- frosting materials.

An everyday example can be observed when squeezing ketchup from a bottle. Initially, the material resists flow, but once movement begins, it flows more readily under continued pressure.

The significance of shear-thinning behaviour in food extrusion systems is that pressure requirements may decrease once flow has been initiated, improving extrusion efficiency.

2.3.5 Shear-Thickening Behaviour

Shear-thickening behaviour occurs when viscosity increases with increasing shear rate. Under higher deformation rates, the material becomes more resistant to flow.

One of the most widely known examples of a shear-thickening material is a suspension of cornstarch and water, commonly referred to as "oobleck." Under slow movement the mixture behaves similarly to a liquid. However, when subjected to sudden impacts or rapid deformation, the material temporarily behaves like a solid.

The phenomenon occurs because suspended particles become increasingly constrained during rapid movement, causing resistance to deformation.

Applications of shear-thickening materials extend beyond food systems and include:

- impact protection materials,
- body armour technologies,
- vibration damping systems,
- advanced protective equipment.

Although shear-thickening behaviour is less common in frosting applications, understanding the phenomenon highlights the diversity of non-Newtonian material behaviour encountered in engineering systems.

2.3.6 Yield Stress Materials

Certain materials do not begin to flow immediately when a force is applied. Instead, a minimum stress level must first be exceeded before movement can occur. This threshold value is known as the yield stress.

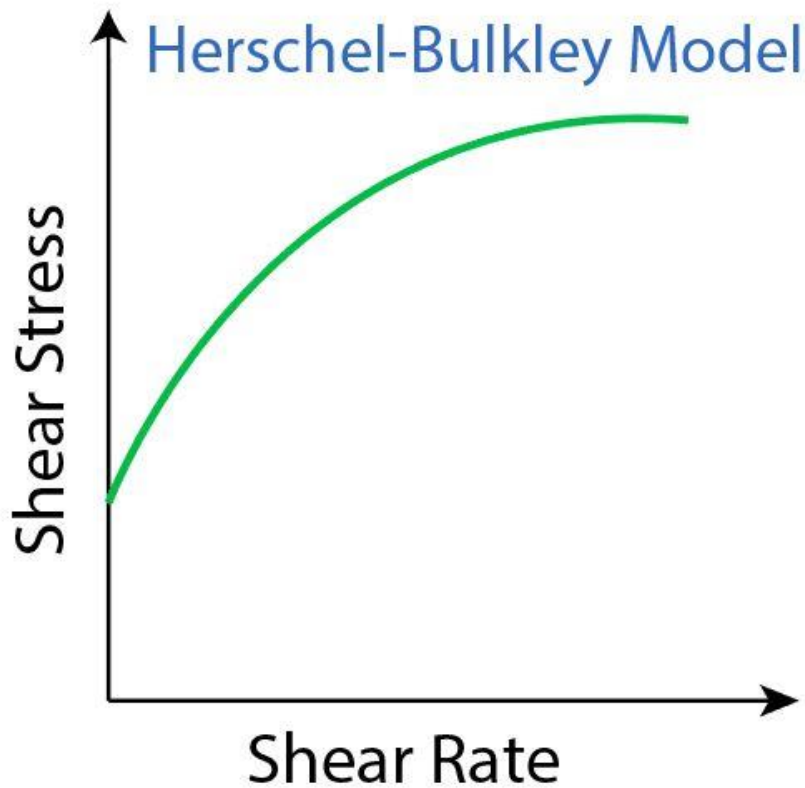
Materials exhibiting yield stress behaviour behave similarly to solids at low stress levels and transition into flowing materials once the yield stress has been exceeded.

Examples include:

- toothpaste,
- mayonnaise,

- clay suspensions,
- many frosting formulations,
- icing materials.

Yield stress is particularly important in automated food deposition because the extrusion system must generate sufficient force to initiate flow before deposition can begin.



2.3.7 Rheological Characteristics of Frosting Materials

Cake frosting materials belong to a broader class of semi-solid food products that exhibit complex rheological behaviour. Unlike simple liquids, frosting materials must satisfy two opposing functional requirements. Firstly, they must be capable of flowing through piping, nozzles, and extrusion systems during application. Secondly, they must retain their shape after deposition to preserve decorative features and maintain the desired surface finish.

These requirements result in frosting materials exhibiting non-Newtonian characteristics, including yield stress behaviour and shear-dependent viscosity. During processing, the material must be subjected to sufficient stress to initiate flow, after which its resistance to deformation may decrease as the shear rate increases. Such behaviour significantly influences the design of

extrusion systems, nozzle geometries, pressure generation mechanisms, and motion control strategies used in automated frosting machines.

The rheological properties of frosting materials affect several aspects of automated deposition, including:

- extrusion pressure requirements,
- flow stability,
- deposition consistency,
- shape retention,
- surface finish quality,
- and overall machine performance.

Consequently, understanding the rheological behaviour of frosting materials is essential for the successful design and operation of automated cake frosting systems.

2.3.8 Common Frosting Materials Used in Cake Decoration

A wide variety of frosting materials are used within the baking and cake-decoration industries. Each material possesses unique rheological characteristics that influence its suitability for automated deposition processes.

Whipped Cream

Whipped cream is a low-density aerated frosting material commonly used for cake finishing and decorative applications. It exhibits relatively low viscosity and can generally be extruded using modest pressure levels. These characteristics make it suitable for dispensing systems that require minimal force generation.

Despite its favorable flow properties, whipped cream possesses limited structural stability and relatively poor shape retention. Decorative features may deform over time, particularly when exposed to elevated temperatures. Consequently, whipped cream is often more suitable for applications requiring smooth surface coverage rather than intricate decorative structures.



Cream Cheese Frosting

Cream cheese frosting provides improved structural stability while maintaining relatively good spreadability. Its rheological properties typically lie between those of whipped cream and stiffer frosting materials. This balance allows it to retain decorative features while remaining sufficiently workable during deposition.

However, cream cheese frosting is sensitive to formulation variations and environmental conditions. Changes in ingredient composition and temperature may significantly influence its viscosity and flow characteristics, potentially affecting extrusion consistency.

Buttercream Frosting

Buttercream frosting is among the most widely used frosting materials in commercial cake decoration. It offers a balance between spreadability, decorative capability, and structural stability. The material can be applied over large cake surfaces while maintaining a smooth appearance and preserving decorative features.

From a rheological perspective, buttercream exhibits relatively high viscosity and yield stress characteristics compared with lower-viscosity frosting materials. These properties provide excellent shape retention but increase the force and pressure requirements associated with automated extrusion systems.



Royal Icing

Royal icing is characterized by high stiffness, strong yield stress behaviour, and excellent shape retention. Once deposited, the material rapidly hardens and effectively maintains its geometry. These characteristics make royal icing suitable for detailed decorative applications, lettering, and intricate piping work.

However, its high resistance to flow increases extrusion pressure requirements and may complicate continuous deposition operations. Automated systems designed for royal icing typically require greater force generation capability than systems intended for softer frosting materials.

Ganache

Ganache is widely used for decorative coatings and smooth surface finishing applications. When appropriately conditioned, it exhibits favorable spreading characteristics and can produce aesthetically appealing surface finishes.

A major challenge associated with ganache is its strong dependence on temperature. Small temperature variations can significantly alter viscosity and flow behaviour, making process control particularly important in automated deposition systems.



Fondant Icing

Fondant is commonly employed as a covering material due to its ability to create smooth and visually appealing cake surfaces. It possesses relatively high stiffness and excellent dimensional stability after application.

Although fondant performs well as a covering material, its limited flowability makes it less suitable for conventional extrusion-based deposition systems without additional conditioning or heating. These characteristics often necessitate alternative processing approaches when automation is required.

Meringue-Based Frostings

Meringue-based frostings are generally lighter and less dense than many traditional frosting formulations. They provide good spreadability and attractive visual appearance while maintaining moderate structural stability.

However, their performance is strongly influenced by environmental conditions such as temperature and humidity. These factors may affect deposition consistency and long-term shape retention during automated frosting operations.

Glaze Coatings

Glaze coatings possess relatively low viscosity and excellent flow characteristics. They can spread uniformly across cake surfaces with minimal applied pressure and are often used to achieve smooth finishing effects.

Despite their favorable flow behaviour, glazes exhibit limited structural strength and poor shape retention. Consequently, they are generally unsuitable for decorative applications that require dimensional stability following deposition.

2.3.9 Factors Affecting Frosting Rheology

The rheological properties of frosting materials are influenced by numerous factors associated with formulation, preparation, and environmental conditions. These factors may significantly alter flow behaviour and therefore affect extrusion performance during automated deposition.

One of the most influential factors is ingredient composition. Variations in sugar concentration, fat content, water content, and stabilizing additives can substantially modify viscosity and yield stress characteristics. Similarly, differences in preparation procedures may alter the internal structure of the frosting and influence its resistance to flow.

Mixing and aeration processes also play a significant role in determining rheological behaviour. The incorporation of air during mixing affects density, texture, and apparent viscosity. Changes in aeration levels may therefore influence extrusion pressure requirements and deposition consistency.

Temperature is another critical parameter. Many frosting materials exhibit strong temperature dependence, with viscosity decreasing as temperature increases. Consequently, environmental conditions and storage temperatures can significantly affect processing performance and final product quality.

Additional factors that may influence frosting rheology include:

- moisture content,
- storage duration,
- ingredient quality,
- formulation variations,
- and handling procedures.

These factors contribute to batch-to-batch variability and present challenges for automated frosting systems that rely on repeatable extrusion performance.

2.3.10 Implications for Automated Frosting Systems

The rheological characteristics of frosting materials have important implications for the design and operation of automated cake frosting machines. Unlike low-viscosity fluids, frosting materials often require significant force generation to initiate and sustain flow. The presence of yield stress and shear-dependent viscosity necessitates careful consideration of pressure generation capability, flow control mechanisms, and nozzle design.

Variations in frosting rheology may also influence deposition quality, resulting in inconsistent layer thickness, surface defects, or interruptions in material flow. Consequently, automated frosting systems must be capable of maintaining stable extrusion performance while accommodating changes in material properties.

The design of the extrusion subsystem must therefore consider:

- pressure generation requirements,
- volumetric flow control,
- nozzle geometry,
- delivery system losses,
- and synchronization between motion and extrusion.

Furthermore, the interaction between material behaviour and machine dynamics highlights the importance of coordinated motion control, structural rigidity, and calibration procedures in achieving consistent frosting application.

The rheological characteristics discussed in this section provide the foundation for understanding the extrusion mechanisms and deposition technologies reviewed in the following section.

2.4 Extrusion Systems for Food Deposition

Food extrusion systems operate by generating pressure within a material reservoir and directing the material through a nozzle or outlet restriction. The generated pressure must overcome several forms of resistance within the flow path, including:

- material yield stress,
- internal friction,
- tubing losses,
- nozzle pressure drop,
- and local flow disturbances.

The ability of an extrusion mechanism to generate and maintain this pressure determines its suitability for automated frosting applications.

2.4.2 Gravity-Fed Extrusion Systems

Gravity-fed systems rely primarily on gravitational forces to drive material flow through a dispensing nozzle.

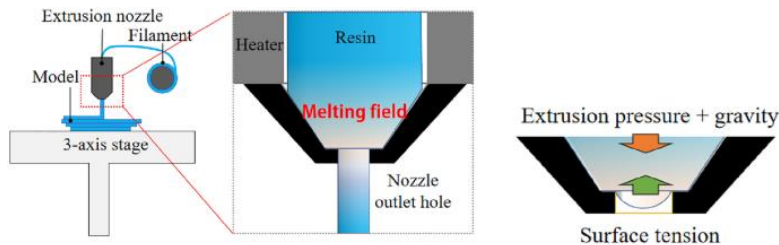
Such systems are simple to construct and require minimal mechanical components. They are commonly used for dispensing low-viscosity materials such as syrups, sauces, and liquid toppings.

Advantages include:

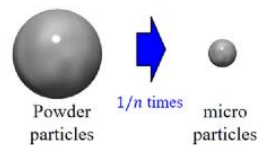
- simple construction,
- low cost,
- low energy requirements,
- minimal maintenance.

However, gravity-fed systems exhibit significant limitations when handling highly viscous materials. Flow rate is strongly influenced by reservoir level, material properties, and outlet geometry. As reservoir contents decrease, pressure available for extrusion also decreases, resulting in inconsistent deposition.

For frosting applications involving medium- to high-viscosity materials, gravity-fed systems generally provide insufficient pressure generation and poor volumetric control.



(a) Relationship between the surface tension and gravity force in the extrusion nozzle.



(b) Concept of moving $1/n$ times size particles.

2.4.3 Screw and Auger Extrusion Systems

Screw-based extrusion systems utilize a rotating auger or helical screw to transport material toward a nozzle while simultaneously generating pressure.

These systems are commonly employed in:

- food processing equipment,
- chocolate dispensing,
- dough extrusion,
- and food printing systems.

The continuous rotational motion of the screw provides continuous material delivery, making auger systems attractive for applications requiring uninterrupted extrusion.

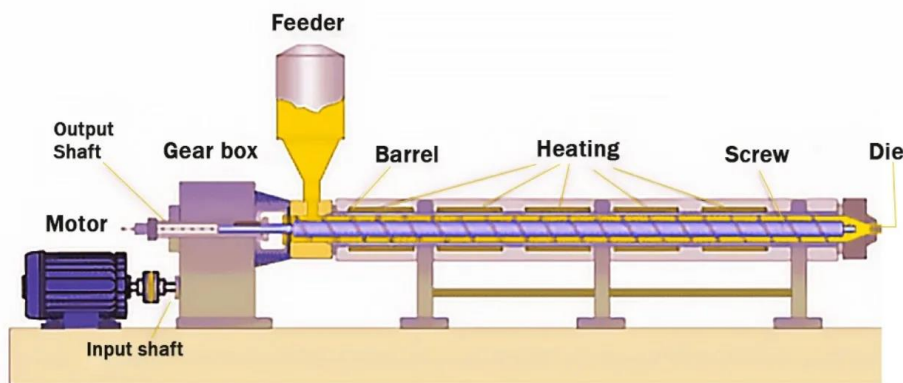
Advantages include:

- continuous operation,
- high throughput capability,
- good pressure generation,
- suitability for large-scale production.

However, several challenges arise when processing frosting materials:

- mechanical shearing may alter material structure,
- cleaning complexity increases,
- material residue may accumulate within screw flights,
- flow pulsation may occur under certain operating conditions.

For decorative frosting applications requiring accurate volumetric control and rapid response to changing deposition commands, screw systems may introduce unnecessary complexity.



2.4.4 Pneumatic Extrusion Systems

Pneumatic extrusion systems generate pressure through compressed air acting on a material reservoir.

The applied air pressure forces material through a nozzle, producing relatively smooth and continuous flow. Pneumatic systems are widely used in food decoration, laboratory dispensing equipment, and additive manufacturing systems involving edible materials.

Advantages include:

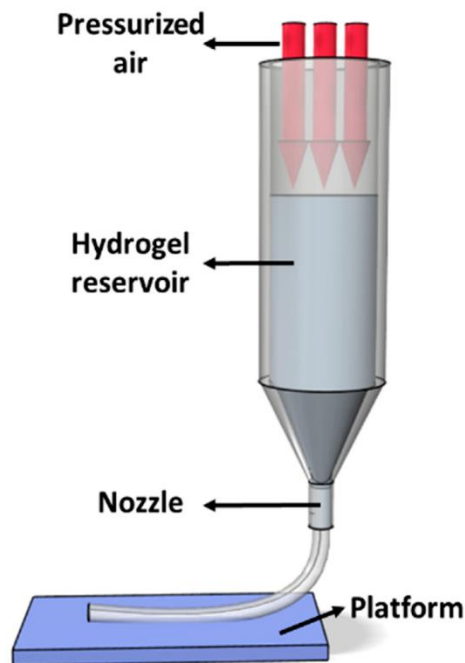
- smooth flow characteristics,
- reduced moving mechanical components,
- relatively simple construction,

- rapid pressure adjustment.

However, pneumatic systems also present several disadvantages:

- requirement for compressed air infrastructure,
- difficulty maintaining precise volumetric output,
- sensitivity to pressure fluctuations,
- delayed response due to material compressibility.

For frosting deposition applications requiring precise synchronization with machine motion, pneumatic systems may experience challenges in maintaining consistent extrusion rates.



2.4.6 Positive Displacement Extrusion Systems

Positive displacement extrusion refers to any extrusion mechanism in which material flow is generated through direct displacement of a known volume.

Examples include:

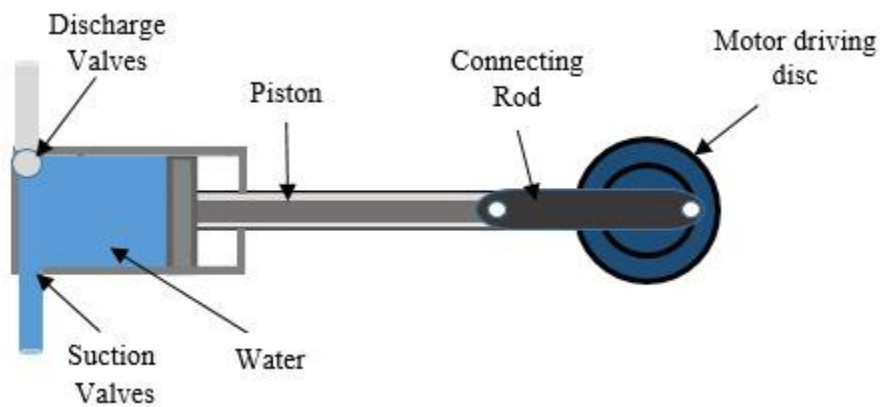
- syringe systems,
- piston systems,
- diaphragm pumps,
- cavity pumps.

Unlike gravity-fed or pressure-regulated systems, positive displacement mechanisms generate flow through mechanical movement rather than relying solely on pressure differences.

A key characteristic of positive displacement systems is that volumetric output is directly related to actuator displacement.

Advantages include:

- high pressure generation capability,
- predictable volumetric flow,
- improved repeatability,
- suitability for highly viscous materials,
- compatibility with automated control systems.



2.4.5 Syringe-Based Extrusion Systems

Syringe extrusion systems are among the most commonly used deposition mechanisms in food printing and automated dispensing applications.

A syringe system consists of:

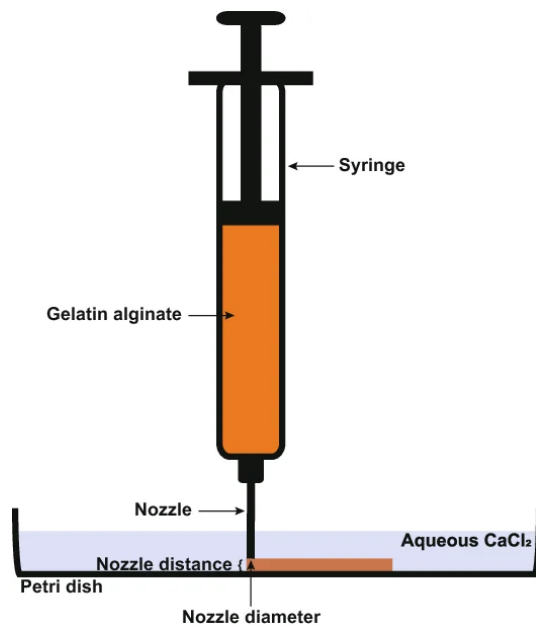
- a cylindrical reservoir,
- a movable plunger,
- an outlet nozzle,
- and an actuation mechanism.

Material flow is generated by advancing the plunger within the reservoir, thereby reducing internal volume and forcing material through the nozzle.

The volumetric displacement produced by syringe systems is directly related to plunger travel, enabling highly predictable material delivery.

Advantages include:

- accurate volumetric control,
- excellent compatibility with viscous materials,
- relatively simple construction,
- easy calibration,
- straightforward cleaning and maintenance.



2.5 Motion Systems Used in Automated Manufacturing

Motion systems form the foundation of automated manufacturing equipment by providing controlled movement of tools, workpieces, or both within a defined workspace. The selection of an appropriate motion system significantly influences machine performance, positional accuracy, structural rigidity, control complexity, manufacturing cost, and ease of implementation. In automated deposition applications such as frosting machines, the motion system determines how accurately the deposition nozzle can be positioned relative to the cake surface and therefore directly affects frosting quality and repeatability.

Modern automated manufacturing systems employ various motion architectures depending on the required workspace, payload capacity, speed, precision, and application requirements. Common motion systems include:

- Cartesian systems,
- SCARA systems,
- Delta robotic systems,
- articulated robotic systems.

Each architecture offers distinct advantages and limitations that must be evaluated against the intended application.

2.5.1 Cartesian Motion Systems

Cartesian motion systems utilize linear movement along mutually perpendicular axes, typically designated as X, Y, and Z. Motion is achieved through independent linear actuators such as lead screws, timing belts, rack-and-pinion drives, or linear motors.

Because movement occurs directly along orthogonal coordinate axes, Cartesian systems provide relatively simple kinematic relationships between commanded positions and physical motion. This simplicity reduces computational requirements and facilitates accurate positioning and control.

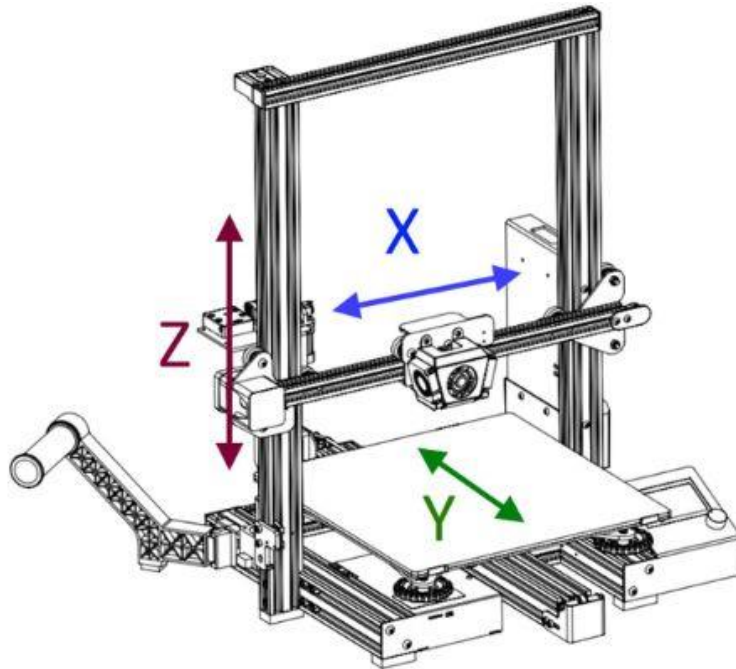
Cartesian systems are widely used in:

- CNC machines,
- 3D printers,
- pick-and-place equipment,
- automated dispensing systems,
- food deposition machinery,
- laboratory automation systems.

Major advantages include:

- high positional accuracy,
- good structural rigidity,
- predictable workspace geometry,
- ease of calibration,
- relatively low implementation cost.

However, Cartesian systems may require larger structural frameworks than some robotic alternatives and can experience increased moving mass depending on the arrangement of the axes.



2.5.2 Gantry-Based Cartesian Systems

Gantry systems represent a specialized implementation of Cartesian motion systems in which one axis spans across the machine structure while supporting one or more additional axes. Gantry configurations are widely used in applications requiring large workspaces, high accessibility, and stable positioning performance.

Two common gantry configurations are employed in industrial automation systems.

Fixed Gantry Systems

In a fixed gantry configuration, the bridge structure remains stationary while the workpiece or supporting platform moves beneath it.

Advantages include:

- high structural rigidity,
- reduced moving mass,
- improved positioning accuracy,
- good vibration resistance.

Disadvantages include:

- larger workspace requirements,
- increased complexity of moving workpiece mechanisms,
- reduced accessibility in some applications.

Fixed gantry systems are commonly used in CNC machining centres and precision manufacturing equipment.



Moving Gantry Systems

In a moving gantry configuration, the bridge structure translates across the machine frame while carrying additional motion axes.

Advantages include:

- efficient workspace utilization,
- improved accessibility,
- simplified workpiece positioning,
- reduced machine footprint,
- straightforward integration of dispensing systems.

Moving gantry systems are widely used in:

- CNC routers,
- laser cutting systems,
- additive manufacturing equipment,
- automated dispensing machines,

- food deposition systems.

For frosting applications, the moving gantry arrangement allows the deposition nozzle to access both the top and side surfaces of the cake while maintaining an unobstructed workspace around the rotary platform.



2.5.3 SCARA Motion Systems

Selective Compliance Assembly Robot Arm (SCARA) systems utilize rotational joints to provide movement primarily within a horizontal plane, together with limited vertical motion. SCARA robots are designed to combine high-speed operation with good positional accuracy and are commonly used in assembly, packaging, and material-handling applications.

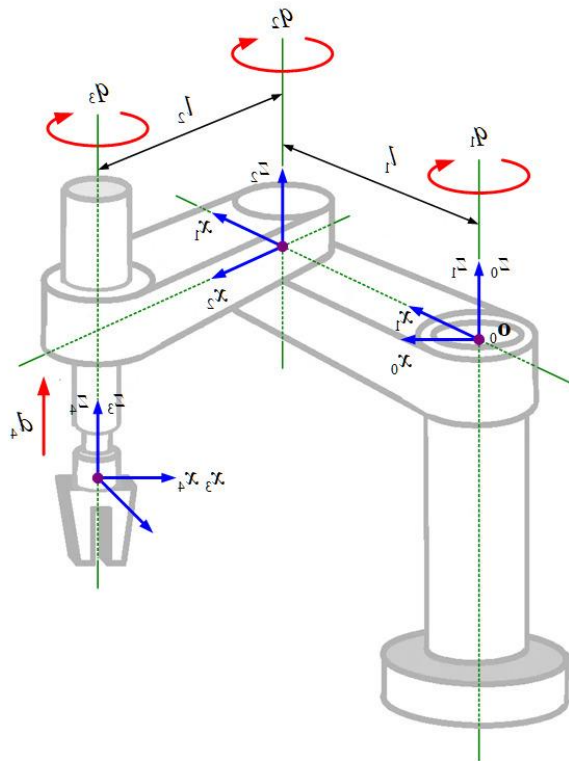
SCARA systems offer several advantages including:

- high operating speed,
- compact workspace utilization,
- relatively low moving mass,
- good repeatability.

However, their kinematic structure introduces increased control complexity compared with Cartesian systems. Positioning requires inverse kinematic calculations, and the achievable workspace is generally less intuitive than that of Cartesian architectures.

SCARA robots are widely used in:

- electronic assembly,
- packaging systems,
- pick-and-place operations,
- laboratory automation.



Although SCARA systems can perform dispensing tasks, their rotational linkage arrangements may complicate synchronization with external rotational mechanisms such as cake turntables.

2.5.4 Delta Robotic Systems

Delta robots are parallel robotic systems consisting of multiple articulated arms connected to a common end effector. Motion is generated through coordinated movement of the parallel linkages.

Delta robots are particularly known for their high-speed positioning capability and low moving mass. These characteristics make them suitable for applications requiring rapid repetitive motion.

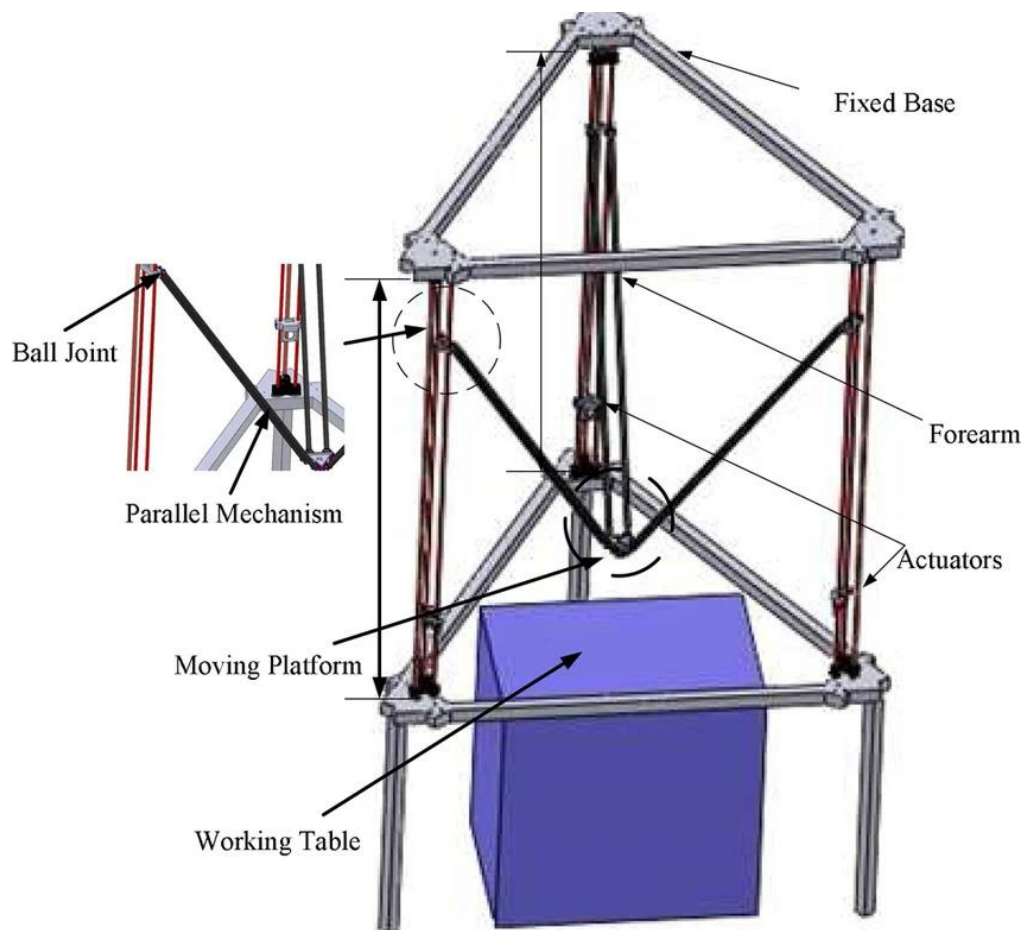
Common applications include:

- food packaging,
- sorting systems,
- pick-and-place operations,
- pharmaceutical handling,
- lightweight assembly processes.

Major advantages include:

- extremely high operating speeds,
- low moving mass,
- excellent acceleration capability,
- high throughput.

However, Delta systems possess more complex kinematic relationships than Cartesian systems and require sophisticated control algorithms and calibration procedures. Their structural behaviour may also become more difficult to analyze when handling varying process loads.



2.6 Rotary Positioning Systems

Rotary positioning systems are widely used in automated manufacturing to provide controlled angular motion of workpieces, tooling systems, and inspection platforms. Unlike linear motion systems that produce translational movement, rotary positioning systems generate rotational motion about a fixed axis. These systems are commonly employed where circumferential processing, indexing operations, or angular positioning is required.

Rotary tables are among the most widely used rotary positioning devices in manufacturing systems. They provide controlled angular displacement and allow a workpiece to be rotated to predetermined positions or continuously rotated during processing operations.



In CNC machining applications, rotary tables provide an additional axis of motion that allows machining operations to be performed on multiple surfaces of a workpiece without repositioning. This capability improves manufacturing accuracy and reduces setup time.

In indexing systems, rotary tables rotate workpieces between processing stations during automated assembly or production operations. The rotational movement is precisely controlled to ensure repeatable positioning at each station.

Rotary positioning systems play an important role in modern food-processing and food-decoration equipment. Rotational motion is frequently used to improve product handling, ensure uniform processing, and facilitate automated decoration operations.

In cake decorating applications, rotating turntables are widely used by professional bakers and decorators to improve frosting consistency and facilitate access to all sides of the cake. Manual cake turntables allow decorators to rotate the cake while maintaining a stationary decorating tool.

Automated cake decorating systems extend this concept by replacing manual rotation with motorized rotary platforms synchronized with automated deposition mechanisms. Such systems enable continuous frosting application around cylindrical cake surfaces and improve repeatability compared to manual methods.

Failure to maintain this synchronization may result in frosting overlap, surface gaps, inconsistent thickness, or poor surface finish.

Consequently, rotary positioning systems have become an important component in modern automated food-decoration and frosting equipment.

2.7 Motion Transmission Systems

Motion systems require mechanisms capable of converting electrical energy into controlled mechanical movement. Motion transmission systems are responsible for transferring and transforming motor output into useful linear or rotational motion. In CNC machines, robotic systems, automated dispensing equipment, and food deposition machines, this process typically involves stepper motors, shaft couplings, lead screw mechanisms, and timing belt drives operating together as an integrated system.

2.7.1 Stepper Motors

Stepper motors are electromechanical devices that convert electrical pulse signals into discrete angular displacements. Unlike conventional DC motors, which rotate continuously when supplied with electrical power, stepper motors rotate through a series of precisely defined angular increments known as steps.

The operating principle of a stepper motor is based on the sequential energization of stator windings, producing magnetic fields that attract and reposition the rotor. By controlling the frequency and sequence of electrical pulses, the position, speed, and direction of rotation can be accurately regulated.

The most common stepper motor types include:

- permanent magnet stepper motors,
- variable reluctance stepper motors,
- hybrid stepper motors.

Among these, hybrid stepper motors are the most widely used in CNC and automation applications due to their favourable combination of positional accuracy, holding torque, and reliability.



Stepper Motor Resolution

The angular displacement produced for each input pulse is referred to as the step angle. A typical hybrid stepper motor possesses 200 steps per revolution, corresponding to a step angle of 1.8° .

The relationship is expressed as:

$$\theta_s = 360^\circ / N$$

where:

Parameter	Description
θ_s	Step angle
N	Number of steps per revolution

Smaller step angles provide higher positioning resolution and improved motion control.

Microstepping

To further increase motion smoothness and positioning resolution, modern motion control systems employ microstepping techniques. Microstepping subdivides each full motor step into smaller increments through proportional control of motor phase currents.

Common microstepping configurations include:

- 1/2 step,
- 1/4 step,
- 1/8 step,

- 1/16 step,
- 1/32 step.

Microstepping improves motion smoothness, reduces vibration, and minimizes acoustic noise. Consequently, it has become a standard feature in CNC machines, 3D printers, and precision automation equipment.

Applications of Stepper Motors

Stepper motors are widely used in:

- CNC machines,
- 3D printers,
- robotic systems,
- automated dispensing equipment,
- laboratory automation systems,
- food processing and deposition machinery.

Their ability to provide accurate open-loop position control makes them particularly suitable for small and medium-scale automation systems where simplicity and cost effectiveness are important considerations.

2.7.2 Shaft Couplings and Power Transmission Interfaces

The rotational motion generated by a stepper motor must be transmitted to the mechanical components responsible for producing useful movement. This transmission commonly occurs through shaft couplings that connect motor shafts to lead screws, pulleys, gear systems, or other rotating elements.

Couplings are therefore critical components within motion transmission systems because they influence torque transfer efficiency, alignment accuracy, vibration behaviour, and overall mechanical reliability.

Rigid Couplers

Rigid couplers establish a direct mechanical connection between two shafts without accommodating misalignment.

Advantages include:

- high torsional stiffness,
- accurate torque transmission,
- simple construction.

However, rigid couplers require precise shaft alignment and may transfer vibration directly between connected components.



Flexible Couplers

Flexible couplers incorporate elastic features that permit limited compensation for angular, parallel, or axial misalignment.

Common flexible coupler types include:

- helical beam couplers,
- jaw couplers,
- Oldham couplers,
- spider couplers.

These designs reduce mechanical stress and vibration while improving system reliability in applications where perfect alignment cannot be guaranteed.



Custom and Manufactured Couplers

Couplers may be produced using a variety of manufacturing methods, including conventional machining processes such as turning and milling, as well as additive manufacturing techniques. Custom-manufactured couplers are often required when commercially available components do not match the dimensional or functional requirements of a particular application.

The use of custom couplers provides several advantages, including:

- adaptation to non-standard shaft dimensions,
- design flexibility,
- integration of application-specific features,
- rapid prototyping and development.

Machined couplers generally offer superior strength, stiffness, and durability, while additively manufactured couplers provide lower manufacturing costs and faster design iteration during prototype development. Consequently, custom couplers are widely employed in research, educational, and prototype automation systems where unique mechanical interfaces are required.

This approach is particularly useful when connecting components such as stepper motor shafts, lead screws, pulleys, or other motion transmission elements that possess differing dimensions or mounting requirements.

2.7.3 Lead Screw Mechanisms

One of the most common methods of converting rotary motion into controlled linear motion is the lead screw mechanism. Lead screws consist of a threaded shaft and mating nut that translate rotational movement into linear displacement.

The mechanism operates on the principle of a helical inclined plane, enabling relatively small motor torques to generate substantial linear forces.

Lead screw systems are extensively used in:

- CNC positioning systems,
- lifting mechanisms,
- extrusion systems,
- automated dispensing equipment,
- robotic motion systems.

Linear Motion Generation

When a lead screw completes one revolution, the mating nut advances by a distance equal to the screw lead.

The linear displacement is given by:

$$L=nP$$

where:

Parameter	Description
L	Linear displacement
n	Number of screw revolutions
P	Lead screw pitch or lead

This relationship enables accurate prediction of linear position from motor rotation.

Mechanical Advantage

A significant advantage of lead screws is their ability to produce large linear forces through mechanical advantage.

Benefits include:

- high force generation,
- excellent positional control,

- suitability for vertical lifting,
- self-locking behaviour in certain designs.

These characteristics make lead screws particularly suitable for extrusion mechanisms and Z-axis positioning systems where substantial linear forces are required.

2.7.4 Timing Belt Systems

While lead screws are commonly used for force generation and precision positioning, timing belts are frequently employed where motion must be transmitted over larger distances or between multiple rotating shafts.

Timing belts utilize molded teeth that engage matching pulley grooves, producing positive mechanical engagement without slip.

This allows precise synchronization of motion while maintaining relatively high operating speeds.

Timing belt systems are widely used in:

- CNC machines,
- 3D printers,
- robotic systems,
- automated manufacturing equipment,
- automated dispensing systems.

Motion Transmission

Timing belts provide:

- high-speed operation,
- low maintenance,
- lightweight construction,
- long travel capability,
- relatively low cost.

These characteristics make them particularly suitable for X-axis and Y-axis positioning systems.

Shaft Synchronization

An important application of timing belts is the synchronization of multiple rotating shafts.

Examples include:

- dual lead screw systems,
- gantry mechanisms,
- multi-axis positioning systems,
- synchronized extrusion systems.

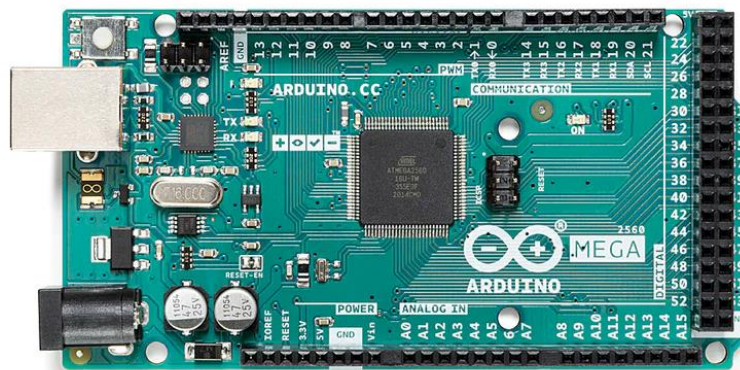
Synchronization improves positional accuracy, reduces mechanical skewing, and ensures uniform force distribution across the mechanism.

For automated frosting equipment, timing belts can be used to synchronize dual lead screws in both vertical lifting systems and syringe-based extrusion mechanisms.

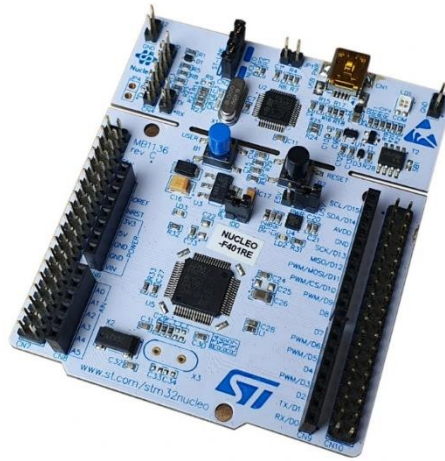
2.8 Controller System

Motion control systems used in automated frosting applications must coordinate multi-axis movement, extrusion control, limit switch monitoring, and user interface communication while maintaining accurate timing for synchronized motion execution. Several embedded controller platforms are commonly used in CNC and robotic systems, including Arduino-based systems, STM32 platforms, Raspberry Pi systems, and ESP32-based controllers.

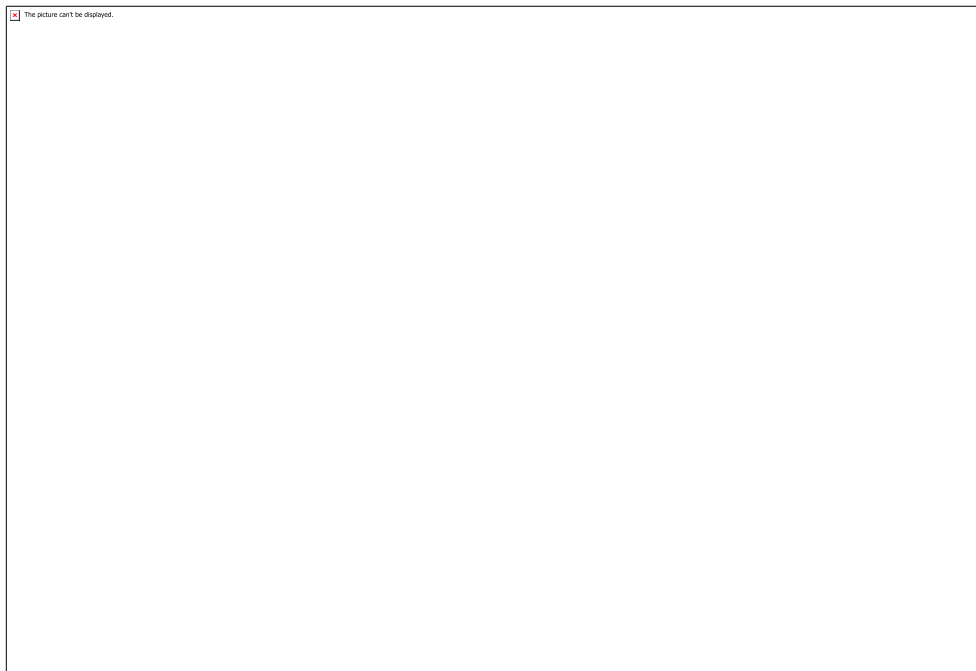
Arduino Mega platforms are widely used in hobby CNC applications due to their low cost and extensive community support. However, their limited processing capability and lower clock speeds may restrict their ability to handle advanced motion synchronization and real-time volumetric calculations in multi-axis systems.



STM32-based controllers provide improved computational capability and are commonly used in industrial embedded systems. These controllers support higher processing speeds and improved peripheral integration but may require more complex firmware development and configuration.

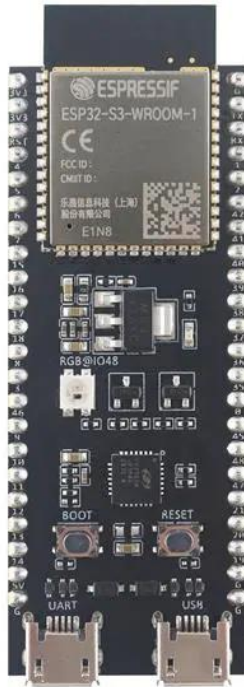


Raspberry Pi systems provide high-level computational capability and support advanced user interfaces and image processing applications. However, Raspberry Pi systems typically operate on non-real-time operating systems, which may introduce timing inconsistencies during precision motion control unless supplemented with dedicated motion-control hardware.



ESP32-based controllers provide a balance between processing capability, wireless communication, peripheral support, and real-time embedded control. The ESP32-S3 specifically incorporates:

- dual-core processing,
- higher clock speeds,
- wireless communication capability,
- compatibility with CNC-oriented firmware platforms



2.10 Challenges Reported in Automated Food Deposition Systems

Automated food deposition systems have become increasingly important in modern food manufacturing due to their ability to improve productivity, consistency, and product quality. Despite these advantages, numerous studies have reported technical challenges associated with the handling and deposition of viscous and semi-solid food materials. The performance of food deposition systems is often influenced by material rheology, machine design, motion synchronization, environmental conditions, and sanitation requirements.

Understanding these challenges is important during the design of automated frosting machines because many of the limitations identified in previous systems directly influence frosting quality, operational reliability, and production efficiency.

2.10.1 Nozzle Clogging and Flow Obstruction

One of the most commonly reported challenges in food extrusion and deposition systems is nozzle clogging. Food materials often contain particulate matter, air pockets, sugar crystals, or fat agglomerations that may accumulate within restricted flow passages.

Nozzle clogging can result in:

- interrupted material flow,
- non-uniform deposition,
- increased extrusion pressure,
- deterioration of surface finish quality,
- and increased cleaning requirements.

The likelihood of clogging generally increases as nozzle diameter decreases. While smaller nozzles improve deposition precision and decorative capability, they simultaneously increase flow resistance and susceptibility to blockage.

Researchers in food extrusion and food-printing systems have therefore emphasized the importance of balancing nozzle geometry with material characteristics to maintain stable flow conditions.

2.10.2 Variations in Frosting Viscosity

A significant challenge in frosting deposition systems is the variability of frosting rheological properties.

The viscosity of frosting materials may change due to:

- temperature fluctuations,
- ingredient composition,
- mixing procedures,
- storage duration,
- and moisture content.

Because frosting materials are typically non-Newtonian, small changes in processing conditions can significantly alter flow behavior.

Variations in viscosity may result in:

- inconsistent extrusion rates,
- fluctuations in deposited layer thickness,
- changes in required extrusion pressure,
- and reduced repeatability between frosting operations.

These variations make it difficult to achieve consistent frosting quality using fixed operating parameters.

For automated systems, the extrusion subsystem must therefore be capable of accommodating a range of material properties while maintaining stable deposition performance.

2.10.3 Synchronization Errors Between Motion and Extrusion

Automated deposition processes require precise coordination between material extrusion and machine movement.

If extrusion flow rate and nozzle motion are not synchronized, defects may occur including:

- excessive frosting accumulation,
- under-deposition,
- surface gaps,
- frosting overlaps,
- and inconsistent layer thickness.

Synchronization challenges become more significant when multiple axes operate simultaneously.

In cake frosting applications, synchronization is required between:

- nozzle translation,
- cake rotation,
- vertical positioning,
- and extrusion flow rate.

Studies involving robotic food decoration systems have shown that motion coordination is critical for achieving smooth and repeatable frosting surfaces.

As machine complexity increases, advanced control strategies become necessary to maintain synchronization among all subsystems.

2.10.4 Structural Deflection and Vibration

Mechanical vibration and structural deflection can negatively affect deposition accuracy.

Sources of vibration may include:

- rapid acceleration and deceleration of moving components,
- belt oscillations,
- motor-induced vibration,
- rotating mechanisms,
- and inadequate structural rigidity.

Even small positional deviations may become visible in frosting applications where smooth surface finishes are required.

Reported consequences include:

- uneven frosting layers,
- dimensional inaccuracies,
- surface waviness,
- and reduced deposition repeatability.

To minimize these effects, machine structures must provide adequate rigidity while maintaining acceptable weight and manufacturability.

Structural design therefore plays a critical role in achieving consistent frosting quality.

2.10.5 Cleaning and Sanitation Requirements

Food-processing equipment must be designed to facilitate cleaning and prevent contamination.

Deposition systems frequently contain:

- reservoirs,
- tubing,
- fittings,
- nozzles,
- and moving components

that come into direct contact with food products.

Poorly designed systems may contain regions where food residues accumulate, creating potential sanitation concerns.

Common design considerations include:

- smooth material-contact surfaces,
- removable food-contact components,
- accessibility for cleaning,
- corrosion resistance,
- and compatibility with food-safe materials.

Cleaning requirements often influence material selection, component geometry, and overall machine architecture.

2.10.6 Food Safety Considerations

Food safety is an essential consideration in automated food handling equipment.

Several factors influence food safety performance, including:

- material compatibility,
- contamination prevention,
- cleaning effectiveness,
- environmental exposure,
- and operator interaction.

Food-contact components are commonly manufactured from materials that:

- resist corrosion,
- do not react with food products,
- are easy to clean,
- and maintain structural integrity during operation.

Designers must also consider methods of reducing contamination risks associated with exposed moving components, lubricants, and inaccessible surfaces.

Compliance with food safety principles is therefore a key requirement in the development of automated frosting systems.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the methodology followed in the modification and improvement of the automated cake frosting machine. The design process incorporated principles from mechatronics, food extrusion systems, fluid mechanics, motion control, and computer numerical control (CNC) system integration. Emphasis was placed on improving frosting deposition consistency, extrusion stability, structural rigidity, and synchronized multi-axis operation when handling cake frosting.

The methodology adopted for the project followed a progressive engineering design approach consisting of:

1. identification of system requirements,
2. analysis of existing limitations,
3. evaluation of alternative design solutions,
4. subsystem selection and justification,
5. engineering calculations,
6. computer-aided design development,
7. system integration planning.

The design methodology also focused on addressing limitations identified in the previous frosting system, particularly in relation to:

- frosting extrusion capability,
- rotational synchronization,
- frosting consistency,
- cake structural stability during operation.

3.2 System Design Requirements

The automated frosting system was required to satisfy several functional and operational requirements associated with frosting deposition on cylindrical cakes.

The primary design requirements included:

- controlled volumetric deposition of buttercream frosting,
- synchronized multi-axis motion control,
- frosting application on both top and side cake surfaces,
- structural rigidity during operation,
- repeatable frosting thickness,
- sufficient extrusion force generation for high-viscosity frosting materials.

The machine was also required to:

- accommodate cakes up to 8 inches in diameter,
- support rotational frosting application,
- minimize frosting deposition inconsistency,
- maintain manageable system cost and complexity.

Additional design considerations included:

- ease of cleaning,
- food-safe material handling,
- modular subsystem integration,
- compatibility with motion control firmware.

3.3 Overall Machine Design Concept

The machine consists of the following primary subsystems:

1. Structural Frame System
2. Cartesian Motion System
3. Theta-Axis Rotary Platform
4. Frosting Extrusion System
5. Electrical Power System
6. Motion Control and Embedded Electronics System
7. Software Control System

The structural frame provides the mechanical foundation of the machine and supports all moving and stationary components.

The Cartesian motion system provides controlled positioning of the frosting nozzle relative to the cake through coordinated movement along the X, Y, and Z axes.

The Theta-axis rotary platform provides controlled rotation of the cake during side-wall frosting operations, enabling continuous frosting deposition around the circumference of the cake.

The frosting extrusion system generates the pressure required to transport frosting from the reservoir to the nozzle and regulates the volumetric flow rate during deposition.

The electrical subsystem provides power distribution, motor actuation, sensor interfacing, and communication between machine components.

The embedded control subsystem coordinates machine operation by controlling motion, extrusion, synchronization, and system monitoring functions.

Custom software executes machine control routines, coordinates subsystem interaction, and provides the logic required for automated frosting operations.

3.3.1 Overall System Architecture

The overall machine architecture follows a Cartesian flying gantry configuration integrated with a rotary cake platform and a positive-displacement frosting extrusion mechanism.

In the selected configuration:

- The X-axis controls horizontal nozzle positioning.
- The Y-axis controls gantry translation across the machine workspace.
- The Z-axis controls vertical positioning of the cake platform.
- The Theta-axis rotates the cake during side-wall frosting.
- The extrusion subsystem delivers frosting to the nozzle under controlled pressure.

During operation, the controller synchronizes nozzle motion, cake rotation, and frosting extrusion to achieve the desired frosting pattern and thickness. This coordinated operation allows frosting to be applied to complex cake surfaces while maintaining consistency and repeatability.

3.3.2 Machine Operating Principle

The frosting process begins by positioning the cake on the rotary platform. The motion system then moves either the cake or frosting nozzle to the required position while the extrusion subsystem delivers frosting through the nozzle.

For top-surface frosting, coordinated X-Y motion is used to generate the desired frosting path.

For side-wall frosting, the Theta-axis rotates the cake while the nozzle position and extrusion rate are synchronized to maintain continuous frosting deposition around the cake circumference.

The integrated operation of these subsystems forms the basis of the automated frosting machine design.

Figure 3.1: Overall Machine Design Concept

(Insert complete machine CAD model showing the Structural Frame, X-Axis, Y-Axis, Z-Axis, Theta-Axis Rotary Platform, Extrusion System, Controller, and Power System. All major subsystems should be clearly labelled.)

3.3 Motion Architecture Selection

Several motion architectures commonly used in robotic positioning and automated manufacturing systems were evaluated during the design process. The architectures considered included:

- SCARA systems,
- Delta robot systems,
- Cartesian gantry systems.

3.3.4 Selection of Cartesian Moving Gantry Architecture

Following evaluation of the available motion architectures, a Cartesian moving gantry configuration was selected for the automated cake frosting machine.

The selection was based on the operational requirements identified in Section 3.2, particularly the need for:

- accurate nozzle positioning,
- repeatable frosting deposition,
- structural rigidity,
- simplified motion control,
- integration with a rotary cake platform,
- compatibility with custom embedded control systems.

Table 3.1: Motion Architecture Comparison

Criterion	SCARA Architecture	Delta Architecture	Cartesian Gantry Architecture
Kinematic Complexity	Moderate to high; requires inverse kinematic calculations for position control.	High; parallel-linkage kinematics require complex motion transformations and calibration.	Low; direct correspondence between axis motion and workspace coordinates simplifies control.
Structural Rigidity	Moderate; rigidity may decrease under varying loads due to articulated joints.	Moderate; stiffness can vary depending on arm configuration and applied loads.	High; linear guide systems provide stable and rigid motion under load.
Workspace Accessibility	Good access within a compact operating envelope.	Good for overhead operations but may be restricted near workspace boundaries.	Excellent; unobstructed access across the entire rectangular workspace.
Motion Accuracy	Good positioning accuracy but dependent on joint calibration and control precision.	High speed but accuracy may be affected by calibration and linkage tolerances.	High positional accuracy and repeatability due to independent linear axes.
Integration with Rotary Platform	Requires coordinated rotational and translational motion control.	Requires complex synchronization between robot motion and cake rotation.	Simple integration due to direct Cartesian positioning and predictable motion paths.
Implementation Cost	Moderate; requires articulated joints and more advanced control systems.	High; complex mechanical structure and calibration requirements increase cost.	Moderate; utilizes standard linear motion components and simpler control architecture.

Criterion	SCARA Architecture	Delta Architecture	Cartesian Gantry Architecture
Suitability for Frosting Applications	Suitable for moderate-speed decorative operations but may face rigidity limitations.	Suitable for high-speed handling tasks but less ideal for precision frosting deposition of viscous materials.	Highly suitable due to accuracy, rigidity, ease of control, and compatibility with rotary cake platforms.

3.5 Motion System Design

Following the selection of a Cartesian moving gantry architecture, the motion system was designed to provide controlled positioning of the frosting nozzle relative to the cake while maintaining sufficient rigidity, repeatability, and synchronization during frosting operations.

The resulting motion system consists of:

- X-axis linear motion subsystem,
- Y-axis gantry translation subsystem,
- Z-axis cake elevation subsystem,
- Theta-axis rotary platform,
- extrusion positioning assembly.

Insert Figure 3.X: Motion System Overview

3.5.1 Motion System Functional Requirements

The motion system was required to satisfy several operational requirements associated with automated frosting deposition.

Table 3.X presents the primary motion system requirements.

Table 3.X Motion System Requirements

Requirement	Description
Positioning Accuracy	Accurate nozzle placement
Repeatability	Consistent frosting deposition
Workspace Coverage	Access to all frosting regions
Synchronization	Coordination with extrusion and rotation
Structural Stability	Reduced vibration and deflection
Ease of Control	Compatibility with custom firmware

3.5.2 X-Axis Design

The X-axis provides horizontal positioning of the frosting nozzle across the width of the cake.

The axis supports the frosting nozzle assembly and enables controlled lateral motion during frosting operations.

The X-axis was designed using:

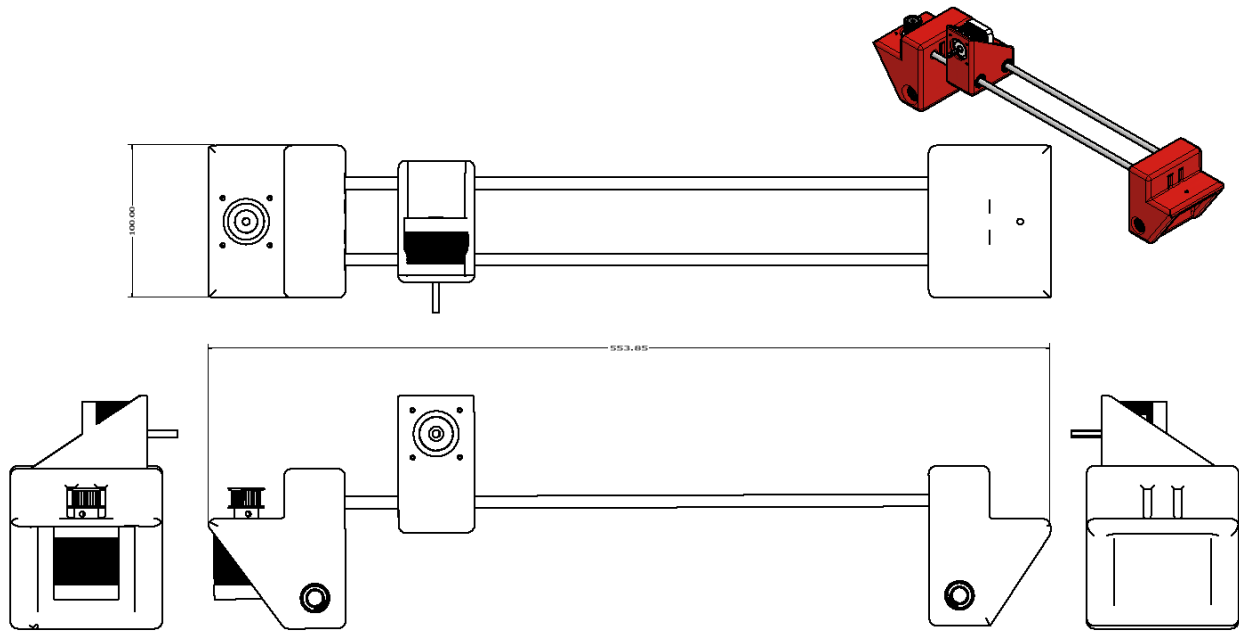
- smooth guide rods,
- linear bearings,
- timing belt transmission,
- stepper motor actuation,
- 3D printed mounting structures.

The use of a belt-driven arrangement provides:

- relatively high positioning speed,
- simplified construction,
- reduced moving mass,
- straightforward integration with the gantry assembly.

The nozzle carriage was designed to support:

- frosting delivery tubing,
- nozzle mounting hardware,
- future interchangeable nozzle assemblies.



X-Axis Travel Determination

The required X-axis travel was determined from:

- maximum cake diameter,
- edge clearance,
- carriage dimensions.

Calculation 3.X: Required X-Axis Travel

- maximum cake diameter = 200 mm
- Tool carriage clearance = 125 mm

$$\text{X-axis travel} = 200 + 125 \times 2 = 450 \text{ mm}$$

3.5.3 Y-Axis Design

The Y-axis provides translational motion of the gantry assembly across the machine frame.

This axis carries:

- X-axis subsystem,
- nozzle carriage,
- frosting delivery assembly.

The Y-axis was designed using:

- dual guide rods,
- timing belt transmission,
- pulley system,
- dual-shaft stepper motor.

The moving gantry configuration was selected because it:

- improves workspace accessibility,
- reduces mechanical complexity,
- simplifies motion control,
- provides direct Cartesian positioning.

Proper belt tensioning was incorporated to minimize backlash and maintain repeatable positioning performance.

Insert Figure 3.X: Y-Axis CAD Assembly

Y-Axis Travel Determination

The Y-axis travel was determined based on:

- cake diameter,
- nozzle access requirements,
- gantry dimensions.

Calculation 3.X: Required Y-Axis Travel

- cake diameter = 200 mm
- nozzle clearance = 100 mm
- Tool carriage clearance = 125 mm

Calculate:

$$Y_{\text{travel}} = 200 + 125 * 2 = 450 \text{ mm}$$

3.5.4 Z-Axis Design

The Z-axis was designed to support and position the cake platform during frosting operations.

The assembly consists of:

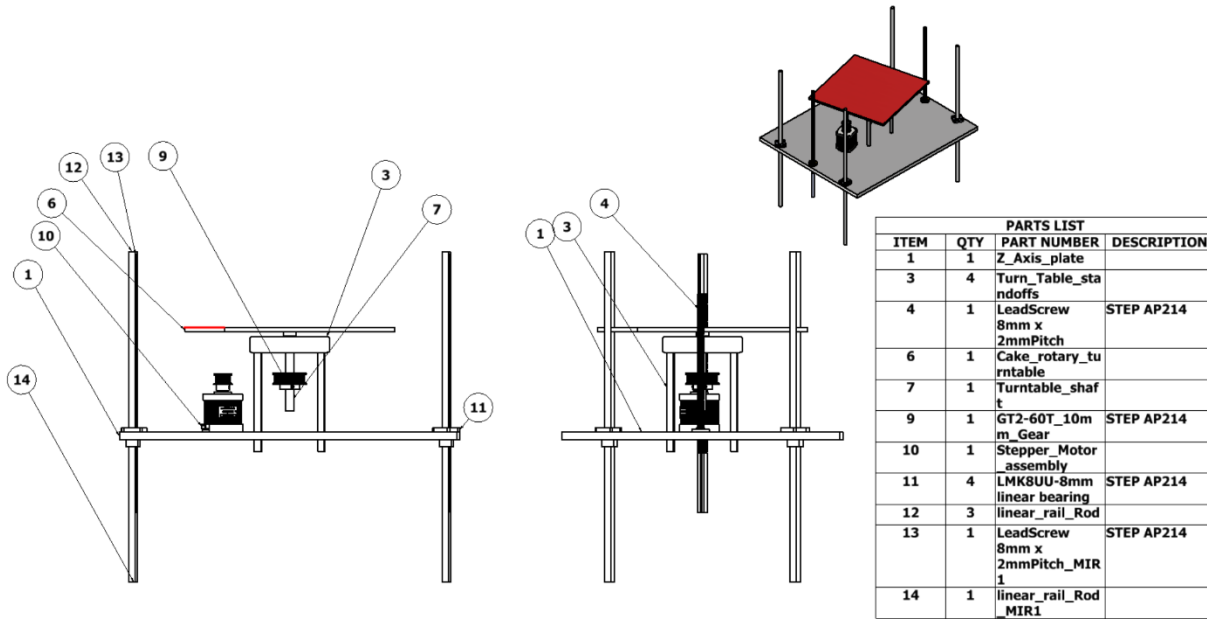
- two stepper motors,
- two lead screws,
- four smooth guide rods,
- platform assembly.

The guide rods provide mechanical support and prevent rotational movement of the platform while the lead screws provide vertical actuation.

The arrangement utilizes:

- two guide rods on the left side,
- two guide rods on the right side,
- one lead screw positioned centrally on each side.

This configuration improves platform stability and minimizes skewing during vertical motion.



Z-Axis Synchronization Method

Several synchronization approaches were evaluated, including:

- independent dual motors,
- mechanically linked lead screws,
- belt-synchronized lead screws.

Independent motor systems provide flexibility but may experience positional mismatch if synchronization is lost.

Mechanically linked systems provide reliable motion but increase mechanical complexity.

A belt-synchronized dual lead screw arrangement was selected because it provides:

- reliable synchronization,
- reduced positional error,
- improved platform stability,
- simplified implementation.

Z-Axis Load Analysis

The Z-axis supports:

- cake mass,
- rotary platform,
- platform structure,
- frosting load.

Z-Axis Load Estimation

- **Guide Rods:** $4 \times 8\text{mm}$ smooth steel rods (approx. 450mm length): 0.7 kg
- **Actuation:** $2 \times \text{T8}$ Lead Screws (approx. 450mm length): 0.3 kg
- **Motors:** $2 \times \text{NEMA 17}$ Stepper Motors: 0.70 kg
- **Platform:** Perspex Base (5mm thick, 300x300mm area): 0.5 kg
- **Hardware:** Linear bearings, lead screw nuts, and synchronization belt: 0.30 kg

Total load = $0.7 + 0.3 + 0.7 + 0.5 + 0.3 = 2.5$ kg

Motor Justification

3.6.1 Structural Design Requirements

The frame was designed to support the combined loading generated by:

- X-axis assembly,
- Y-axis gantry assembly,
- Z-axis lifting assembly,
- Theta-axis rotary platform,
- extrusion subsystem,
- electrical components.

Table 3.X summarizes the primary structural design requirements.

Table 3.X: Structural Frame Requirements

Requirement	Design Objective
Rigidity	Minimize deflection during operation
Stability	Prevent excessive vibration
Load Capacity	Support all machine subsystems
Manufacturability	Allow fabrication using available tools
Maintainability	Permit easy access to components
Expandability	Support future modifications
Cost Effectiveness	Minimize fabrication cost

3.6.2 Evaluation of Structural Frame Alternatives

Several structural construction approaches were evaluated before selecting the final frame design.

Welded Steel Frame

Welded steel structures are commonly used in industrial machinery due to their high stiffness and load carrying capability.

Aluminium Extrusion Frame

Aluminium extrusion systems provide modular construction using standardized profiles and fasteners.

Sheet Metal Construction

Sheet metal structures provide lightweight machine construction and compact packaging.

Stainless Steel Frame

Stainless steel structures are commonly used in food-processing applications due to their corrosion resistance and sanitary properties.

Table 3.X: Structural Frame Alternative Comparison

Criterion	Welded Steel	Aluminium Extrusion	Sheet Metal	Stainless Steel
Rigidity	High	Moderate	Moderate	High
Weight	High	Low	Low	High
Fabrication Simplicity	Moderate	High	Moderate	Moderate
Cost	Low	Moderate	Moderate	High
Expandability	Low	High	Moderate	Low
Modification Capability	Low	High	Moderate	Low

3.6.3 Selection of Frame Structure

Following evaluation of the available alternatives, a welded steel frame was selected for the proposed machine.

The selection was based on:

- availability of fabrication equipment,
- material accessibility,
- structural rigidity,
- ability to support motion components,
- ability to support extrusion loads,
- cost effectiveness.

The welded construction method also provided a robust support platform for the motion system and reduced the likelihood of structural movement during frosting operations.

3.6.4 Frame Configuration

The selected frame was designed as a cubic support structure consisting of welded mild steel members.

The frame provides mounting interfaces for:

- Y-axis guide system,
- Z-axis lifting mechanism,
- extrusion subsystem,
- electrical enclosure,
- control electronics.

The overall frame dimensions were selected to provide sufficient workspace while maintaining a compact machine footprint.

Table 3.X: Frame Dimensions

Parameter	Value
Length	450 mm
Width	450 mm
Height	450 mm
Material	Mild Steel
Section Type	Square Hollow Section
Wall Thickness	30 mm

Figure 3.X: Structural Frame CAD Model

Figure 3.X: Frame Orthographic Views

3.6.7 Structural Deflection Considerations

Excessive frame deflection can affect frosting quality by introducing positioning errors into the motion system.

The frame design therefore sought to:

- minimize unsupported spans,
- distribute loads uniformly,
- provide adequate joint stiffness,
- reduce vibration.

The selected frame geometry improves rigidity by utilizing closed-loop rectangular structures and welded corner joints.

A detailed finite element analysis was not conducted; however, the selected frame dimensions and material properties were considered adequate for the anticipated operational loading conditions.

3.7 Theta-Axis Rotary Platform Design

The frosting of cylindrical cake surfaces requires controlled rotational motion to ensure continuous frosting deposition around the circumference of the cake.

Requirements

Requirement	Purpose
Cake Support	Carry cake weight during frosting
Controlled Rotation	Maintain repeatable rotational motion
Motion Synchronization	Coordinate rotation with frosting deposition
Positioning Accuracy	Ensure repeatable angular positioning
Stability	Minimize wobble and vibration
Load Capacity	Accommodate different cake sizes
Manufacturability	Enable practical fabrication

The platform was designed to accommodate cakes with diameters of up to 8 inches while maintaining sufficient stability throughout the frosting process.

3.7.2 Evaluation of Rotary Platform Alternatives

Several rotary drive approaches were evaluated during the design process.

Direct Drive Rotation

In a direct-drive arrangement, the rotary platform is mounted directly to the motor shaft.

Belt-Driven Rotation

Belt-driven systems utilize timing belts and pulleys to transmit rotational motion between the motor and the rotating platform.

Gear-Driven Rotation

Gear-driven systems utilize spur gears, bevel gears, or gear reduction systems to transmit motion.

3.7.3 Selection of Rotary Drive Mechanism

Following evaluation of the available alternatives, a custom belt-driven rotary platform was selected for the proposed machine.

The selection was based on:

- simplicity of fabrication,
- ease of motor placement,
- ability to increase output torque,
- compatibility with the Z-axis platform,
- reduced mechanical complexity compared with gear systems.

The selected configuration allows the motor to be positioned independently of the rotating platform while providing sufficient torque for cake rotation.

Table 3.X: Rotary Platform Selection Matrix

Criterion	Direct Drive	Belt Drive	Gear Drive
Simplicity	High	Moderate	Low
Torque Capability	Low	High	High
Cost	Low	Moderate	High
Fabrication Ease	High	High	Low
Maintenance	High	Moderate	Moderate
Final Selection		✓	

3.7.4 Rotary Motion Analysis

For continuous side-wall frosting, rotational motion must be coordinated with frosting deposition.

The relationship between tangential velocity and angular velocity is expressed as:

$$v=r\omega$$

Where:

Symbol	Description
v	Tangential velocity
r	Cake radius
ω	Angular velocity

This relationship forms the basis for synchronization between rotational motion and frosting application.

Calculation 3.7: Maximum Cake Radius

- maximum cake diameter = 200mm

Maximum cake radius = 100 mm

Calculation 3.8: Required Rotational Speed

- nozzle feed rate,
- cake radius.

Using:

$$v=r\omega$$

- angular velocity =
- rotational speed in revolutions per minute (RPM) =

3.7.5 Rotary Platform Structural Design

The rotary platform consists of:

- rotating cake platform,
- support bearings,
- drive pulley,
- timing belt,
- stepper motor,
- mounting structure.

The platform was designed to:

- support cake loading,
- maintain rotational alignment,
- reduce wobble,
- provide a stable frosting surface.

Figure 3.X

Theta-Axis CAD Assembly

3.7.6 Bearing Selection Considerations

The rotating platform requires bearing support to maintain smooth motion while carrying the cake load.

The bearing arrangement was required to:

- support axial loading,
- support radial loading,
- reduce rotational friction,
- maintain alignment.

Several approaches may be considered including:

- deep groove ball bearings,
- lazy-susan bearings,
- thrust bearings,
- custom bearing assemblies.

3.7.7 Torque Requirement Analysis

The drive motor must generate sufficient torque to:

- accelerate the platform,
- overcome bearing friction,
- rotate the cake,
- maintain rotational motion during frosting.

Calculation 3.9: Platform Inertia

Determine:

- cake mass = 1 kg
- platform mass,
- platform radius.

Estimate:

$$J = \frac{1}{2}mr^2$$

where:

- J = rotational inertia,
- m = rotating mass,
- r = radius.

Calculation 3.10: Required Rotary Torque

Using:

- rotational inertia,
- angular acceleration,

$$T = J\alpha$$

where:

- T = torque,
- J = inertia,
- α = angular acceleration.

3.7.8 Integration with the Motion System

The Theta-axis platform is integrated directly with the Z-axis lifting assembly.

This configuration enables:

- vertical positioning of the cake,
- rotational frosting operations,
- synchronized motion with nozzle movement,
- coordinated frosting deposition.

The integration of the Theta-axis significantly improves frosting coverage on cylindrical surfaces and addresses limitations identified in previous frosting machine designs where rotational synchronization was not adequately controlled.

3.8 Frosting Material Selection and Rheological Analysis

The performance of an automated frosting machine is strongly influenced by the physical and rheological properties of the frosting material being deposited. The selected frosting affects:

- extrusion force requirements,
- pressure generation,
- flow behaviour,
- deposition consistency,
- surface finish quality,
- frosting thickness control,
- nozzle performance.

Consequently, frosting selection formed an important stage of the design process because the chosen material establishes the operational requirements of the extrusion subsystem.

Several frosting materials commonly used in cake decoration were evaluated before selecting the design frosting for the proposed machine.

3.8.1 Frosting Selection Criteria

The frosting materials were evaluated using criteria relevant to automated deposition systems.

Table 3.X: Frosting Selection Criteria

Criterion	Description
Ease of Extrusion	Force required to achieve flow
Shape Retention	Ability to maintain deposited geometry
Continuous Deposition Suitability	Ability to support uninterrupted flow
Temperature Sensitivity	Stability under environmental conditions
Surface Finish Quality	Ability to produce smooth frosting surfaces
Commercial Relevance	Frequency of use in cake decoration
Automation Compatibility	Suitability for machine application

3.8.3 Frosting Material Comparison

The evaluated frosting materials were compared using the previously established criteria.

Table 3.X: Frosting Material Comparison

Property	Whipped Cream	Cream Cheese	Royal Icing	Ganache	Buttercream
Ease of Extrusion	High	Moderate	Low	Moderate	Moderate
Shape Retention	Low	Moderate	High	Moderate	High

Property	Whipped Cream	Cream Cheese	Royal Icing	Ganache	Buttercream
Temperature Sensitivity	High	Moderate	Low	High	Moderate
Continuous Deposition Suitability	Moderate	Moderate	Low	Moderate	High
Surface Finish Quality	Moderate	Moderate	Moderate	High	High
Commercial Usage	Moderate	Moderate	Moderate	Moderate	High

3.8.4 Selection of Design Frosting

Following evaluation of the available frosting materials, buttercream frosting was selected as the primary design material for the proposed machine.

The selection was based on:

- widespread commercial usage,
- suitability for automated frosting operations,
- ability to maintain decorative features,
- compatibility with continuous deposition,
- acceptable surface finish characteristics.

The selected frosting represents one of the more challenging materials to automate due to its relatively high viscosity and non-Newtonian flow characteristics.

Consequently, further rheological analysis was required to establish the design requirements of the extrusion subsystem.

3.8.5 Rheological Behaviour of the Selected Frosting

Unlike Newtonian fluids, whose viscosity remains constant during flow, frosting materials generally exhibit non-Newtonian behaviour.

Buttercream demonstrates:

- yield stress behaviour,
- shear-thinning characteristics,
- time-dependent flow response.

As a result, the frosting does not begin flowing immediately when pressure is applied. A minimum stress level must first be exceeded before flow occurs.

Once flow begins, the apparent viscosity decreases as the shear rate increases.

These characteristics significantly influence:

- extrusion pressure requirements,
- nozzle performance,
- flow stability,
- deposition quality.

3.8.6 Herschel-Bulkley Flow Model

The flow behaviour of many frosting materials may be represented using the Herschel-Bulkley model.

$$\tau = \tau_y + K\dot{\gamma}^n$$

Where:

Symbol	Description
τ	Shear stress
τ_y	Yield stress
K	Consistency index
$\dot{\gamma}$	Shear rate
n	Flow behaviour index

This model accounts for:

- yield stress,
- viscosity variation,
- shear-thinning behaviour.

The relationship provides a useful approximation for estimating extrusion pressure requirements during frosting deposition.

3.8.7 Implications for Extrusion System Design

The rheological analysis establishes several requirements for the frosting delivery system.

The extrusion subsystem must be capable of:

- generating sufficient pressure to initiate flow,
- maintaining continuous frosting delivery,
- overcoming nozzle resistance,
- overcoming tubing losses,
- minimizing flow interruptions,
- producing repeatable volumetric output.

The frosting delivery mechanism must therefore provide significantly greater force generation than would be required for low-viscosity materials such as whipped cream.

These requirements formed the basis for the evaluation and selection of suitable extrusion mechanisms presented in the following section.

3.9 Extrusion System Design and Selection

The extrusion subsystem is responsible for delivering frosting from the storage reservoir to the frosting nozzle in a controlled and repeatable manner.

Unlike conventional liquid dispensing applications, cake frosting materials exhibit relatively high viscosity and non-Newtonian flow behaviour. Consequently, the extrusion system must generate sufficient force to initiate and maintain frosting flow while ensuring consistent deposition throughout the frosting process.

The extrusion subsystem therefore serves as the interface between the selected frosting material and the motion system.

The design objectives of the extrusion subsystem were:

- controlled volumetric delivery,
- repeatable frosting deposition,
- compatibility with high-viscosity frosting,
- continuous frosting flow,
- ease of cleaning,
- ease of fabrication,
- integration with automated control.

3.9.1 Extrusion System Requirements

The rheological analysis presented in Section 3.8 established several performance requirements for the frosting delivery system.

The extrusion mechanism was required to:

- generate sufficient pressure to overcome frosting yield stress,
- maintain continuous flow during operation,
- provide controllable flow rate,
- support frosting deposition over extended periods,
- integrate with the machine control system.

Table 3.X: Extrusion System Requirements

Requirement	Purpose
Pressure Generation	Initiate frosting flow
Flow Stability	Maintain consistent deposition
Volumetric Control	Control frosting thickness
Reliability	Reduce interruptions
Cleanability	Facilitate maintenance
Manufacturability	Support practical fabrication

3.9.2 Evaluation of Extrusion Alternatives

Several extrusion approaches commonly used in food processing and dispensing systems were evaluated.

Gravity-Fed Extrusion

Gravity-fed systems rely primarily on gravitational force to generate material flow.

Because frosting viscosity can be significantly higher than whipped cream or liquid toppings, gravity-fed systems were considered unsuitable for the proposed machine.

Syringe Extrusion

Syringe extrusion systems utilize a piston to force material from a cylindrical reservoir through a nozzle.

Syringe extrusion is widely used in food printing and paste extrusion applications due to its ability to provide controlled material deposition.

Auger Extrusion

Auger systems utilize a rotating screw to transport material toward the nozzle.

The possibility of altering frosting consistency through mechanical shearing reduced the attractiveness of this approach.

Piston Pump Systems

Piston pumps utilize reciprocating motion to generate flow.

3.9.3 Selection of Lead Screw Driven Syringe Extrusion

Following evaluation of the available approaches, a lead screw driven syringe extrusion system was selected.

The selected approach combines:

- direct volumetric displacement,
- high force generation capability,
- simple construction,
- compatibility with automated control.

The selected configuration utilizes a large-capacity cylindrical reservoir with a mechanically driven plunger.

Rotary motion generated by a stepper motor is converted into linear motion using lead screws, producing controlled displacement of the plunger and subsequent extrusion of frosting.

Table 3.X: Extrusion Method Comparison

Criterion	Gravity Feed	Auger	Piston Pump	Syringe
Pressure Capability	Low	Moderate	High	High
Flow Control	Low	Moderate	High	High
Cleaning Ease	High	Low	Moderate	High
Mechanical Complexity	Low	High	High	Moderate
Frosting Compatibility	Low	Moderate	High	High
Selected				✓

3.9.4 Extrusion System Architecture

The selected extrusion subsystem consists of:

- cylindrical frosting reservoir,
- plunger assembly,
- dual lead screw drive,
- synchronization belt,
- stepper motor,
- frosting delivery hose,
- nozzle assembly.

The system operates by translating the plunger downward through the reservoir, forcing frosting toward the outlet and subsequently through the frosting nozzle.

Figure 3.X: Extrusion System Architecture

3.9.5 Reservoir Design Considerations

The frosting reservoir serves as the storage chamber for frosting prior to extrusion.

The reservoir was designed to:

- maximize storage capacity,
- provide smooth internal surfaces,
- support food-safe operation,
- simplify fabrication.

The selected reservoir consists of a cylindrical chamber fabricated from PPR pipe.

Table 3.X: Reservoir Parameters

Parameter	Value
Internal Diameter	106 mm
Material	PPR
Orientation	Vertical
Extrusion Method	Bottom Discharge

3.9.6 Plunger Design

The plunger functions as the force transmitting element between the lead screw drive mechanism and the frosting material.

The plunger was designed to:

- maintain sealing within the reservoir,
- distribute force uniformly,
- prevent frosting bypass,
- support connection to the drive mechanism.

The plunger incorporates a central mounting interface that allows attachment to the moving crossbar driven by the lead screws.

Custom Wood Plunger Design

3.9.7 Lead Screw Actuation System

The extrusion system utilizes dual lead screws to generate the force required for frosting extrusion.

The lead screws perform two functions:

- conversion of motor rotation into linear motion,
- maintenance of plunger alignment.

The use of dual screws reduces:

- plunger skewing,
- uneven loading,
- reservoir binding.

Synchronization between both screws is achieved using a timing belt.

Calculation 3.11: Reservoir Cross-Sectional Area

The cross-sectional area of the reservoir is determined using:

$$A=\pi(D/2)^2$$

Where:

- D = reservoir diameter

Using:

$$D=106 \text{ mm}$$

$$\text{Area} = 8824 \text{ square mm}$$

3.9.8 Extrusion Pressure Requirements

The total pressure required for frosting extrusion consists of:

- frosting yield stress,
- tubing losses,
- nozzle losses,
- local flow losses.

Table 3.X: Sources of Pressure Loss

Source	Description
Yield Stress	Pressure required to initiate flow
Tubing Losses	Friction losses
Nozzle Restriction	Nozzle pressure drop
Local Losses	Fittings and transitions

Thus:

$$P_{\text{total}} = P_{\text{yield}} + P_{\text{tube}} + P_{\text{nozzle}} + P_{\text{loss}}$$

Calculation 3.12: Estimated Extrusion Pressure

3.9.9 Required Plunger Force

The force required to drive the frosting through the delivery system is determined by:

$$F = P \times A$$

Where:

Symbol	Description
F	Required force
P	Extrusion pressure
A	Reservoir area

Calculation 3.13: Required Plunger Force

- reservoir area,
- extrusion pressure.

$$F =$$

3.9.10 Motor Torque Requirement

The stepper motor must generate sufficient torque to produce the required plunger force through the lead screw mechanism.

Calculation 3.14: Lead Screw Torque Requirement

Using:

- lead screw pitch,
- lead screw efficiency,
- required plunger force,

$$T = F \times p / 2\pi\eta$$

Where:

- T = torque,
- F = force,
- p = lead screw pitch,
- η = efficiency.

3.9.11 Implications for Nozzle Design

The extrusion analysis demonstrates that nozzle geometry directly affects:

- pressure requirements,
- frosting flow behaviour,
- deposition quality,
- extrusion force.

Consequently, nozzle design forms an important component of the frosting delivery system and is considered separately in the following section.

3.10 Frosting Nozzle Design

The frosting nozzle forms the final stage of the frosting delivery system and directly influences frosting deposition quality, surface finish, layer thickness, and overall machine performance.

While the extrusion subsystem is responsible for generating the force required to move frosting through the delivery system, the nozzle governs how the frosting exits the system and is applied onto the cake surface.

The nozzle design therefore required consideration of:

- frosting flow characteristics,
- pressure losses,
- frosting layer thickness,
- surface finish quality,
- nozzle manufacturability,
- cleaning requirements,
- compatibility with the selected extrusion system.

Since the machine is intended for automated frosting of cylindrical cakes, the nozzle must support continuous frosting deposition while maintaining a consistent frosting profile throughout machine operation.

3.10.1 Functional Requirements of the Nozzle

The nozzle was required to satisfy the following functional requirements:

Requirement	Purpose
Continuous Flow	Prevent interruption of frosting deposition
Uniform Frosting Layer	Maintain consistent frosting thickness
Low Flow Resistance	Reduce extrusion pressure requirements
Food Safety	Allow contact with edible materials
Ease of Cleaning	Facilitate maintenance and sanitation
Manufacturability	Allow economical fabrication
Compatibility	Integrate with extrusion tubing and motion system

The nozzle must additionally minimize frosting accumulation around the outlet region and prevent undesirable flow disturbances that could affect frosting appearance.

3.10.2 Frosting Nozzle Design Alternatives

Several nozzle configurations commonly used in frosting, food extrusion, and dispensing applications were considered.

Straight Circular Nozzle

The straight circular nozzle consists of a cylindrical outlet with a constant internal diameter.

Tapered Conical Nozzle

A tapered nozzle gradually reduces the internal flow area from inlet to outlet.

Slot Nozzle

Slot nozzles produce a continuous ribbon of frosting through a narrow rectangular opening.

Decorative Frosting Nozzles

Decorative nozzles are widely used in manual cake decoration and incorporate specialized outlet geometries.

Because the primary objective of this project is automated frosting coverage rather than decorative piping, decorative nozzle geometries were not considered as the primary frosting nozzle.

3.10.3 Nozzle Selection

Following evaluation of the available alternatives, a custom frosting nozzle was selected for the proposed system.

The selection was based on the need to:

- maintain consistent frosting deposition,
- minimize extrusion pressure losses,
- support automated frosting application,
- improve frosting surface quality,
- integrate with the selected syringe extrusion system.

The final nozzle geometry was developed specifically for the requirements of the frosting machine rather than adopting a standard commercial piping tip.

3.10.4 Custom Nozzle Development

A custom nozzle was designed to improve frosting deposition performance during automated operation.

The development process considered:

- frosting rheology,
- nozzle outlet geometry,
- flow distribution,
- pressure loss minimization,
- manufacturability using additive manufacturing.

The nozzle was designed using CAD software and fabricated using 3D printing techniques to allow rapid prototyping and design iteration.

The custom design enables future modification of:

- outlet dimensions,
- internal flow passages,
- frosting layer width,
- frosting thickness characteristics.

This flexibility provides significant advantages during experimental optimization.

3.10.5 Nozzle Design Parameters

The nozzle design was influenced by several geometric parameters.

Parameter	Influence on Performance
Outlet Diameter	Frosting thickness and pressure drop
Inlet Diameter	Flow transition characteristics
Nozzle Length	Flow stabilization
Exit Geometry	Frosting profile

Parameter	Influence on Performance
Internal Surface Finish	Flow resistance
Outlet Area	Deposition width

These parameters directly influence both frosting quality and extrusion force requirements.

3.10.7 Nozzle Design Calculations

Calculation 3.X: Nozzle Outlet Area

Determine nozzle outlet area using:

$$A = \pi(D/2)^2$$

Where:

- A = nozzle outlet area
- D = nozzle outlet diameter

This calculation is used to estimate:

- frosting discharge area,
- deposition width,
- flow velocity.

Calculation 3.X: Frosting Exit Velocity

Using the continuity relationship:

$$Q = AV$$

- frosting exit velocity,
- volumetric flow rate relationship.

Where:

Parameter	Description
Q	Volumetric flow rate
A	Outlet area
V	Exit velocity

Calculation 3.X: Estimated Nozzle Pressure Drop

3.10.8 Nozzle Design

1. **Nozzle Design 1 CAD**
2. **Nozzle Design 2 CAD**
3. Selected design
4. Nozzle outlet dimensions
5. Nozzle inlet dimensions
6. Manufacturing method (3D printed, machined, etc.)
7. Material used
8. Whether the nozzle is intended for:
 - side-wall frosting,
 - top-surface frosting,
 - or both.

3.11 Motion Control System Design

Unlike conventional CNC systems that rely on G-code interpretation, the proposed frosting machine utilizes custom firmware developed specifically for automated cake frosting operations. This approach allows the implementation of application-specific motion routines tailored to frosting deposition requirements while providing greater flexibility for future system development.

The motion control system was therefore required to coordinate:

- X-axis nozzle positioning,
- Y-axis gantry translation,
- Z-axis cake elevation,
- Theta-axis cake rotation,
- syringe extrusion control,
- homing procedures,
- limit switch monitoring,

- machine sequencing,
- communication interfaces.

The controller additionally provides synchronization between frosting deposition and machine movement to ensure consistent frosting thickness and surface coverage.

3.11.1 Motion Control Requirements

Table 3.17: Motion Control Requirements

Requirement	Purpose
Multi-Axis Coordination	Simultaneous control of X, Y, Z and Theta axes
Real-Time Operation	Accurate timing of motion commands
Extrusion Synchronization	Coordinate frosting flow with machine motion
Sensor Integration	Monitor limit switches and machine status
Expandability	Support future hardware additions
Communication Capability	Allow user interaction and configuration
Custom Motion Logic	Implement frosting-specific movement routines
Reliability	Maintain repeatable operation

The selected controller was required to satisfy these requirements while maintaining low system complexity and manageable implementation cost.

3.11.2 Controller Platform Evaluation

Several embedded control platforms commonly used in automation and motion control applications were evaluated.

The evaluation focused on:

- processing capability,
- real-time performance,
- development flexibility,
- communication capability,
- scalability,
- implementation cost.

Arduino-Based Controllers

Arduino platforms are widely used in educational and hobby automation projects due to their simplicity and extensive support community.

STM32-Based Controllers

STM32 microcontrollers are widely used in industrial and embedded applications requiring higher computational performance.

Raspberry Pi Platforms

Raspberry Pi systems provide substantial computational capability and support advanced user interfaces.

ESP32-S3 Controllers

The ESP32-S3 combines relatively high processing capability with integrated wireless communication and extensive peripheral support.

Table 3.18: Controller Platform Comparison

Criterion	Arduino	STM32	Raspberry Pi	ESP32-S3
Processing Capability	Low	High	Very High	High
Real-Time Performance	Moderate	High	Moderate	High
Wireless Connectivity	Limited	Limited	High	High
GPIO Availability	Moderate	High	Moderate	High
Development Complexity	Low	High	Moderate	Moderate
Cost	Low	Moderate	High	Low
Expandability	Moderate	High	High	High

3.11.3 Selection of ESP32-S3 Controller

Based on the evaluation criteria, the ESP32-S3 was selected as the primary controller for the frosting machine.

The selection was influenced by:

- dual-core processing capability,
- integrated wireless communication,
- sufficient processing resources for coordinated motion control,
- extensive peripheral support,
- low implementation cost,
- compatibility with PlatformIO development tools,
- support for future machine expansion.

3.11.4 Software Development Environment

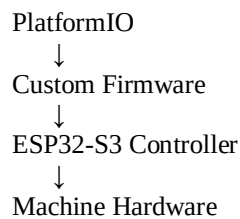
Unlike conventional CNC implementations that rely on preconfigured motion-control firmware, the proposed system utilizes custom-developed firmware specifically tailored to automated cake frosting operations.

This approach provides greater flexibility for implementing:

- custom frosting routines,
- application-specific motion profiles,
- coordinated extrusion control,
- future machine upgrades,
- experimental frosting strategies.

The use of custom firmware additionally allows direct integration of machine-specific operating logic without the constraints associated with generalized CNC firmware platforms.

Figure 3.19: Firmware Development Environment



3.11.5 Software Architecture

To simplify development and improve maintainability, the firmware was divided into several functional modules.

Each module performs a specific function while interacting with the remaining system components through a structured software architecture.

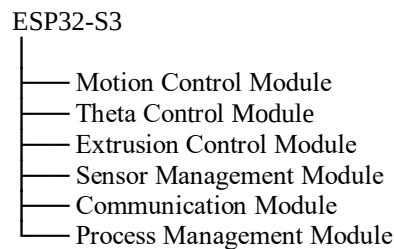
Table 3.19: Software Modules

Module	Function
Motion Control Module	X, Y and Z axis movement
Theta Control Module	Rotary platform control
Extrusion Control Module	Syringe actuation
Sensor Management Module	Limit switch monitoring
Communication Module	Wireless and serial communication
Process Management Module	Frosting sequence execution

Module	Function
Safety Management Module	Fault detection and protection

This modular approach simplifies debugging, future modifications, and software expansion.

Figure 3.20: Software Architecture Block Diagram



3.11.6 Motion Coordination Strategy

The frosting process requires coordinated operation of multiple machine axes and the extrusion subsystem.

Unlike conventional positioning systems where each axis may operate independently, frosting deposition requires synchronization between machine movement and frosting flow to achieve consistent layer thickness and surface finish.

The motion coordination strategy was therefore designed to maintain synchronization between:

Table 3.20: Motion Coordination Functions

Motion Combination	Purpose
X + Y Motion	Top surface frosting
Theta + Z Motion	Side-wall frosting
Extrusion + Motion	Consistent frosting deposition
Homing + Motion	Repeatable machine positioning
Theta + Extrusion	Continuous cylindrical frosting

For example, during side-wall frosting operations:

- the Theta-axis rotates the cake,
- the Z-axis adjusts nozzle elevation,
- the extrusion subsystem delivers frosting,
- all movements occur simultaneously.

The controller therefore schedules and coordinates these actions to maintain a consistent frosting layer around the cake circumference.

3.11.7 Sensor Integration

The motion control system incorporates sensors to improve operational reliability and machine safety.

The sensors provide information required for:

- axis homing,
- travel limit detection,
- machine initialization,
- fault prevention.

Potential sensor types include:

- mechanical limit switches,
- optical sensors,
- inductive proximity sensors,
- Hall-effect sensors.

Table 3.21: Sensor Functions

Sensor	Function
X-Axis Limit Switch	Travel reference
Y-Axis Limit Switch	Travel reference
Z-Axis Limit Switch	Vertical homing
Theta Reference Sensor	Rotational positioning
Emergency Stop Input	Safety shutdown

Later

1. Actual homing sensor type used.
2. Number of limit switches installed.
3. Whether a Theta-axis home sensor is implemented.
4. Whether an emergency stop circuit is included.

3.11.8 Communication and User Interaction

The ESP32-S3 provides integrated wireless communication capability that can support future user interaction and machine monitoring functions.

Potential communication interfaces include:

- USB serial communication,
- Wi-Fi connectivity,
- Bluetooth communication,
- web-based user interfaces.

These interfaces may be used for:

- machine configuration,
- parameter adjustment,
- status monitoring,
- firmware updates,
- diagnostic functions.

The communication subsystem was designed to remain expandable to support future development of advanced user interfaces and remote monitoring capabilities.

3.12 Electrical Power System Design

The electrical power system was designed to provide reliable and stable operation of all machine subsystems. The frosting machine incorporates multiple electrical devices operating at different voltage and current requirements, including stepper motors, motor drivers, embedded controllers, sensors, and communication peripherals.

The electrical power system was therefore designed to support:

- motion actuators,
- extrusion actuation,
- embedded control electronics,
- sensors and limit switches,
- communication devices,
- future expansion modules.

The design additionally considered:

- electrical safety,
- voltage regulation,
- power distribution,
- current capacity,
- ease of maintenance.

3.12.1 Electrical Power Requirements

The electrical requirements of the frosting machine were derived from the selected mechanical and control subsystems.

Major electrical loads include:

Table 3.22: Major Electrical Loads

Subsystem	Function
X-Axis Motor	Nozzle positioning
Y-Axis Motor	Gantry translation
Z-Axis Motors	Vertical platform movement
Theta-Axis Motor	Cake rotation
Syringe Extrusion Motor	Frosting extrusion
ESP32-S3 Controller	System control
Sensors	Position monitoring
Communication Devices	User interaction

The power architecture must therefore support both high-power electromechanical devices and low-power electronic circuits.

3.12.2 Power Architecture Alternatives

Several power-distribution approaches were evaluated.

The evaluation considered:

- implementation complexity,
- electrical compatibility,
- expandability,
- system reliability.

Single-Voltage Architecture

Single-voltage systems utilize one voltage source for all machine components.

Dual-Voltage Architecture

Dual-voltage systems separate actuator power and logic-level power supplies.

Multi-Domain Power Architecture

Multi-domain architectures provide dedicated voltage levels for different categories of devices.

Table 3.23: Power Architecture Comparison

Criterion	Single Voltage	Dual Voltage	Multi-Domain
Simplicity	High	Moderate	Low
Flexibility	Low	Moderate	High
Device Compatibility	Low	Moderate	High
Expandability	Low	Moderate	High
Noise Isolation	Low	Moderate	High
Future Development	Low	Moderate	High

Following evaluation, the multi-domain architecture was selected.

3.12.3 Selected Power Architecture

A multi-domain power system was adopted to accommodate the electrical requirements of the machine.

The selected architecture separates motion power from low-voltage electronics, improving system stability and reducing the likelihood of electrical interference between subsystems.

Table 3.24: Selected Voltage Domains

Voltage Domain	Purpose
24 V	Stepper motors and motion actuation
5 V	Peripheral devices and auxiliary electronics
3.3 V	ESP32-S3 controller logic

This architecture allows each subsystem to operate at its recommended voltage while maintaining a centralized power distribution structure.

Figure 3.21: Electrical Power Architecture

3.12.4 Power Supply Selection

The primary power supply was selected to provide sufficient electrical capacity for simultaneous operation of all machine subsystems.

The power supply must satisfy:

- motor current requirements,
- controller power consumption,
- sensor power requirements,
- startup current demands,
- future expansion allowances.

Calculation 3.X: Estimated System Power Requirement

- motor current ratings,
- driver requirements,
- controller consumption,
- sensor loads.

$$P=VI$$

Where:

Parameter	Description
P	Electrical power
V	Supply voltage
I	Current

use to justify power supply sizing.

Calculation 3.X: Total Current Requirement

1. Power supply voltage.
2. Power supply current rating.
3. Power supply wattage.
4. Whether a single or multiple power supplies are used.

3.12.5 Voltage Regulation

Voltage regulation is required to supply stable operating voltages to the controller and peripheral devices.

Because the primary power domain operates at 24 V, lower-voltage devices require regulated conversion stages.

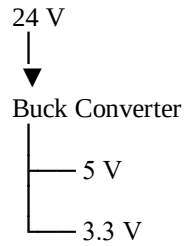
The power system therefore incorporates DC-DC voltage regulators to generate:

- 5 V power,
- 3.3 V logic supply.

The voltage regulation subsystem was designed to:

- maintain voltage stability,
- improve electrical efficiency,
- reduce heat generation,
- protect sensitive electronics.

Figure 3.22: Voltage Regulation Scheme



3.12.6 Motor Driver Power Distribution

The motion system utilizes stepper motor drivers that receive power directly from the primary voltage domain.

The drivers provide:

- current regulation,
- microstepping capability,
- motor protection,
- controlled power delivery.

The power distribution system was designed to provide stable supply voltage to each driver while minimizing voltage drop and electrical noise.

1. Driver model.
2. Driver operating voltage.
3. Driver current setting.
4. Microstepping configuration.

These values will later support driver sizing justification.

3.12.7 Electrical Protection Considerations

Electrical protection measures were incorporated to improve operational reliability and reduce the likelihood of component damage.

Protection strategies may include:

- input fuses,
- reverse polarity protection,
- emergency stop circuitry,

- overcurrent protection,
- voltage regulation protection.

The protection system helps safeguard both the machine hardware and operator during operation.

Table 3.25: Electrical Protection Features

Protection Feature	Purpose
Fuse Protection	Overcurrent protection
Voltage Regulation	Stable supply voltages
Emergency Stop	Immediate machine shutdown
Grounding	Electrical safety
Connector Isolation	Reduce accidental shorts

3.12.8 PCB-Based Power Distribution

To simplify wiring and improve maintainability, power distribution was integrated into the custom PCB developed for the frosting machine.

The PCB serves as the central electrical interface and provides:

- power routing,
- controller integration,
- motor driver connections,
- sensor interfaces,
- expansion capability.

Integrating power distribution into the PCB reduces wiring complexity and improves system organization compared to point-to-point wiring approaches.

Figure 3.23: PCB Power Distribution Layout

3.13 Custom Component Design

In addition to commercially available mechanical and electrical components, several custom-designed components were developed to satisfy the unique functional requirements of the automated frosting machine. These components were necessary because commercially available

alternatives either did not meet the dimensional constraints of the machine or could not provide the required functionality for frosting operations.

The custom components were developed using computer-aided design (CAD) software and manufactured using appropriate fabrication techniques including additive manufacturing, machining, and printed circuit board fabrication.

The objectives of the custom component design process were to:

- improve subsystem integration,
- reduce assembly complexity,
- enhance structural rigidity,
- accommodate unique machine geometries,
- simplify maintenance,
- reduce manufacturing cost,
- support future design modifications.

The major custom-designed components include:

- motor-to-lead screw couplers,
- X-axis mounting assemblies,
- extrusion subsystem support structure,
- frosting nozzle assemblies,
- printed circuit board (PCB),
- sensor mounting brackets,
- cable management components.

Table 3.26: Custom Designed Components

Component	Function
Motor Coupler	Connect motor shaft to lead screw
X-Axis Mounts	Support nozzle carriage assembly
Extrusion Frame	Support syringe extrusion mechanism
Frosting Nozzles	Control frosting deposition
PCB	Electrical integration
Sensor Brackets	Position switches and sensors
Cable Management Components	Organize electrical wiring

The design methodology for each custom component is discussed in the following sections.

3.13.1 Motor-to-Lead Screw Coupler Design

Several machine subsystems required the transmission of torque between stepper motors and lead screws. However, the motor shafts and lead screws possessed different diameters, necessitating the development of custom couplers.

The couplers were designed to:

- transmit torque reliably,
- maintain shaft alignment,
- minimize backlash,
- simplify assembly,
- accommodate manufacturing tolerances.

The specific application involved connecting:

- 5 mm motor shafts,
- 8 mm lead screws.

Commercial couplers were evaluated but custom designs were developed to improve compatibility with the machine geometry and manufacturing constraints.

Design Requirements

Table 3.27: Coupler Design Requirements

Requirement	Purpose
Torque Transmission	Drive lead screws
Shaft Alignment	Reduce vibration
Mechanical Strength	Prevent failure
Manufacturability	Simplify production
Compactness	Fit available space

The couplers were modeled in CAD software and manufactured using additive manufacturing techniques.

Figure 3.24: Motor-to-Lead Screw Coupler CAD Model

- motor side interface,
- lead screw side interface,
- fastening mechanism.

Calculation 3.X: Coupler Torque Verification

Inputs:

- motor torque,
- shaft diameter,
- material properties.

Outputs:

- torsional stress,
- safety factor.

3.13.2 X-Axis Mounting System Design

The X-axis assembly required several custom mounting components to support:

- guide rods,
- belt drive components,
- frosting nozzle carriage,
- sensor mounting locations.

Commercial mounting solutions were not readily compatible with the selected machine geometry and therefore custom brackets were developed.

The mounting components were designed to:

- maintain alignment,
- reduce assembly complexity,
- support repeatable motion,
- minimize structural deflection.

Figure 3.25: X-Axis Mounting Components

- carriage assembly,
- guide rod supports,
- belt attachment features.

Later

1. Number of printed components.
2. Material used.
3. Printer model.
4. Layer height and infill percentage.

3.13.3 Extrusion System Structural Components

The syringe extrusion mechanism required several custom structural components that were not available commercially.

These components include:

- lead screw support brackets,
- motor mounting plates,
- crossbar assembly,
- syringe support structure,
- bearing supports.

The components were designed to maintain alignment of the extrusion subsystem while allowing smooth piston motion during frosting extrusion.

Design Objectives

- maintain piston alignment,
- reduce lead screw skewing,
- distribute extrusion loads,
- simplify maintenance.

Figure 3.26: Extrusion Subsystem Structural Components

- syringe body,
- moving plunger,
- lead screws,
- support frame,
- motor mounting locations.

3.13.8 Additive Manufacturing Considerations

Many of the custom machine components were fabricated using additive manufacturing techniques.

The use of 3D printing provided several advantages:

- rapid prototyping,
- reduced fabrication cost,
- simplified production of complex geometries,
- rapid design iteration.

Components potentially produced through additive manufacturing include:

- couplers,
- nozzle assemblies,
- mounting brackets,
- sensor supports,
- cable management components.

Table 3.29: Potential 3D Printed Components

Component	Purpose
Couplers	Shaft connection
Nozzles	Frosting deposition
Brackets	Mechanical support
Sensor Mounts	Sensor positioning
Cable Guides	Wiring organization

3.14 Fabrication Methodology

Following completion of the design phase, the proposed frosting machine was fabricated and assembled to validate the selected mechanical, electrical, and control system designs. The fabrication process transformed the computer-aided design models into a functional prototype capable of performing automated frosting operations.

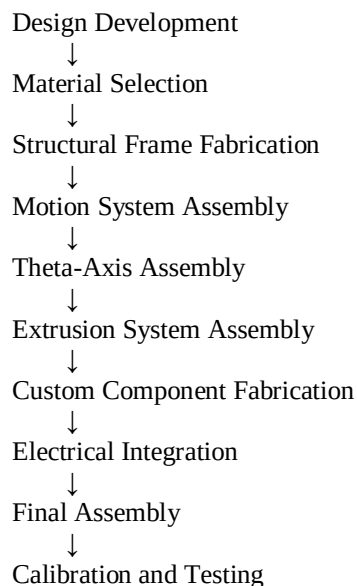
The fabrication methodology was developed to:

- translate the design into a physical machine,
- ensure proper subsystem alignment,
- maintain structural rigidity,
- facilitate maintenance and future modifications,
- simplify assembly procedures,
- provide a platform for testing and validation.

The fabrication process was divided into the following major stages:

1. Structural frame fabrication,
2. Motion system fabrication and assembly,
3. Theta-axis fabrication and assembly,
4. Extrusion system fabrication and assembly,
5. Fabrication of custom-designed components,
6. Electrical subsystem integration,
7. Final machine assembly and alignment.

Figure 3.xx: Overall Fabrication Methodology Flowchart



3.14.1 Manufacturing Methods Employed

The machine incorporated multiple manufacturing techniques selected according to component function, cost, available resources, and fabrication complexity.

Table 3.xx Manufacturing Methods Used

Manufacturing Method	Application
Welding	Structural frame fabrication
Drilling	Mounting holes and fastener locations
Cutting and Grinding	Frame member preparation
3D Printing	Custom brackets, couplers, nozzle mounts,
CNC Cutting	Perspex parts
Mechanical Fastening	Subsystem assembly
PCB Fabrication	Control electronics
Soldering	Electrical interconnections
Woodwork	Plunger mechanism
Plumbing	Nozzle Fabrication

The use of multiple fabrication techniques enabled the integration of custom-designed components while minimizing manufacturing cost.

3.14.2 Structural Frame Fabrication

The structural frame was fabricated as the primary supporting structure of the machine.

The frame supports:

- X-axis assembly,
- Y-axis gantry,
- Z-axis mechanism,
- extrusion subsystem,

- electrical components,
- Theta-axis rotary platform.

The frame was constructed using welded mild steel members selected for their availability, strength, and ease of fabrication.

Fabrication Procedure

The frame fabrication process consisted of:

1. Measurement and marking of frame members.
2. Cutting members to required lengths.
3. Initial positioning and alignment.
4. Tack welding.
5. Dimensional verification.
6. Final welding.
7. Grinding and finishing.
8. Drilling of mounting locations.

The frame was fabricated as a rigid cubic structure designed to minimize vibration and positional deviation during frosting operations.

Figure 3.xx: Structural Frame Fabrication Stages

photographs showing:

- material preparation,
- cutting operations,
- tack welding,
- completed welded frame.

3.14.3 Motion System Fabrication

Following completion of the structural frame, the Cartesian motion system was assembled.

The motion system consists of:

- X-axis assembly,
- Y-axis assembly,
- Z-axis assembly.

The fabrication sequence followed the order of increasing subsystem dependency.

3.14.3.1 Z-Axis Fabrication and Assembly

The Z-axis was assembled first because it provides support for the Theta-axis cake platform.

The assembly incorporates:

- dual lead screws,
- dual stepper motors,
- guide rods,
- moving support platform.

Assembly Procedure

1. Installation of guide rods.
2. Mounting of lead screws.
3. Motor installation.
4. Coupler installation.
5. Platform assembly.
6. Alignment verification.

The guide rods provide positional stability while the lead screws generate controlled vertical movement.

Figure 3.xx: Z-Axis Assembly

- lead screws,
- guide rods,
- motors,
- moving platform.

Later

Parameter	Value
-----------	-------

Lead Screw Diameter	____
---------------------	------

Lead Screw Pitch	____
------------------	------

Parameter	Value
Lead Screw Length	____
Guide Rod Diameter	____

3.14.3.2 Y-Axis Fabrication and Assembly

The Y-axis assembly provides gantry translation across the machine frame.

The assembly incorporates:

- linear guide rods,
- timing belt transmission,
- corner pulleys,
- dual-shaft stepper motor.

Assembly Procedure

1. Installation of support rods.
2. Pulley mounting.
3. Motor installation.
4. Belt installation.
5. Belt tensioning.
6. Alignment verification.

Proper belt tension was maintained to reduce backlash and improve positioning accuracy.

Figure 3.xx: Y-Axis Assembly

- rods,
- belt,
- pulleys,
- motor.

Later

Parameter	Value
-----------	-------

Rod Diameter	___
--------------	-----

Rod Spacing	___
-------------	-----

Belt Type	___
-----------	-----

Pulley Teeth	___
--------------	-----

3.14.3.3 X-Axis Fabrication and Assembly

The X-axis supports the frosting nozzle assembly and provides horizontal positioning during frosting operations.

The assembly incorporates:

- guide rods,
- timing belt transmission,
- nozzle carriage,
- custom 3D-printed mounting brackets.

Assembly Procedure

1. Fabrication of custom mounts.
2. Installation of guide rods.
3. Installation of timing belt.
4. Assembly of nozzle carriage.
5. Alignment verification.

The use of additive manufacturing enabled rapid production of custom mounting interfaces while reducing machining requirements.

Figure 3.xx: X-Axis Assembly

- carriage,
- guide rods,
- nozzle mount,
- belt drive.

Later

Parameter	Value
Filament Material	____
Printer Model	____
Layer Height	____
Infill Percentage	____

3.14.4 Theta-Axis Fabrication and Assembly

The Theta-axis rotary platform was fabricated to provide rotational motion of the cake during frosting operations.

The assembly was mounted on the moving Z-axis platform to enable simultaneous vertical positioning and rotational movement.

Assembly Procedure

1. Fabrication of platform support.
2. Installation of bearings.
3. Motor mounting.
4. Transmission assembly.
5. Platform alignment.
6. Rotation verification.

The completed assembly was integrated with the control system to allow synchronized operation with the Cartesian motion axes.

Figure 3.xx: Theta-Axis Assembly

- rotating platform,
- bearings,
- motor,
- transmission mechanism.

Later

Parameter	Value
Drive Type	Direct / Belt / Gear
Motor Type	_____
Platform Diameter	_____
Bearing Type	_____
Reduction Ratio	_____

3.14.5 Extrusion System Fabrication

The extrusion subsystem represents one of the primary functional elements of the machine and was fabricated to provide controlled frosting delivery.

The extrusion system consists of:

- frosting reservoir,
- syringe chamber,
- plunger assembly,
- dual lead screws,
- synchronization belt,
- stepper motor,
- delivery hose,
- nozzle assembly.

Table 3.xx Extrusion Components

Component	Function
Syringe Reservoir	Frosting storage
Plunger	Material displacement
Lead Screws	Linear actuation
Timing Belt	Synchronization

Component	Function
Stepper Motor	Motion generation
Hose	Material transport
Nozzle	Frosting deposition

Fabrication Procedure

1. Reservoir fabrication.
2. Plunger fabrication.
3. Lead screw installation.
4. Synchronization belt installation.
5. Crossbar assembly.
6. Hose installation.
7. Nozzle installation.
8. Functional verification.

The syringe reservoir was fabricated using a 106 mm internal diameter PPR pipe selected for availability and ease of fabrication.

The plunger assembly was connected to a synchronized dual lead screw mechanism capable of generating the force required for frosting extrusion.

Figure 3.xx: Exploded View of Extrusion System

- reservoir,
- plunger,
- lead screws,
- belt,
- motor,
- hose,
- nozzle.

Figure 3.xx: Assembled Extrusion System

3.14.6 Fabrication of Custom Components

Several machine components could not be sourced commercially and were therefore designed and fabricated specifically for the project.

The custom-designed components include:

- motor-to-lead screw couplers,
- X-axis mounting brackets,
- nozzle mounting interfaces,
- plunger assembly,
- extrusion crossbar,
- PCB mounting brackets,
- sensor mounts.

These components were primarily produced using additive manufacturing due to its flexibility and rapid prototyping capability.

Figure 3.xx: Custom Fabricated Components

- CAD models,
- printed components,
- installed components.

Table 3.xx Custom Components Manufactured

Component	Manufacturing Method
5 mm to 8 mm Coupler	Lathe Work
X-Axis Mount	3D Printing
Sensor Mount	3D Printing
Nozzle Holder	3D Printing
Plunger Assembly	Woodwork
PCB Mount	

3.14.6.1 Fabrication Challenges Encountered

This will discuss:

- alignment challenges,
- lead screw synchronization issues,
- coupler failures or redesigns,
- print tolerances,
- frame squareness issues,
- wiring constraints,
- component availability limitations.

3.15 Electrical Integration and PCB Development

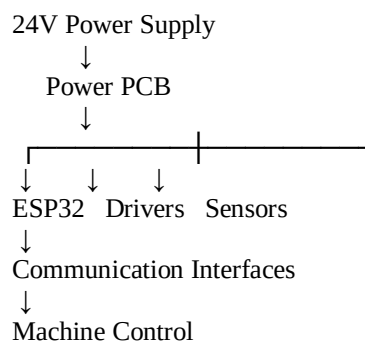
3.15 Introduction

Following fabrication and assembly of the mechanical subsystems, the electrical system was integrated to provide power distribution, motion control, sensor interfacing, and communication functionality.

The electrical system integrates:

- power conversion circuitry,
- motion control electronics,
- motor drivers,
- sensors,
- communication interfaces,
- peripheral expansion connections.

Figure 3.xx: Overall Electrical System Architecture



3.15.1 Electrical System Requirements

The electrical system was required to support all machine subsystems while maintaining stable operation during simultaneous motion and extrusion activities.

Table 3.xx Electrical System Requirements

Requirement	Purpose
Motion Control	Control of X, Y, Z and Theta axes
Extrusion Control	Actuation of syringe extrusion system
Sensor Integration	Limit switches and monitoring
Power Distribution	Supply electrical power to all subsystems
Expandability	Future system upgrades
Communication	Firmware updates and machine interaction
Reliability	Stable machine operation

The electrical design was therefore developed around a modular architecture that separates power distribution, control logic, and actuation subsystems.

3.15.2 Custom PCB Development

To simplify wiring and improve subsystem integration, a custom printed circuit board (PCB) was developed.

The PCB serves as the central electrical interface of the machine.

Its functions include:

- mounting of the ESP32-S3 controller,
- power distribution,
- motor driver interfacing,
- sensor connections,
- communication interfaces,
- future expansion support.

The use of a custom PCB significantly reduced wiring complexity compared to point-to-point wiring approaches.

Table 3.xx PCB Functional Requirements

Requirement	Purpose
Controller Interface	ESP32-S3 integration
Driver Interface	Stepper motor control
Sensor Inputs	Switch and sensor monitoring
Power Distribution	Voltage routing
Expansion Headers	Future subsystem integration
Signal Routing	Organized system interconnection

3.16 Embedded Software Development and Control Implementation

3.16 Introduction

The automated frosting machine requires coordinated operation of multiple mechanical and electrical subsystems to achieve controlled frosting deposition. These subsystems include the Cartesian motion platform, Theta-axis rotary platform, syringe-based extrusion system, sensors, and communication interfaces.

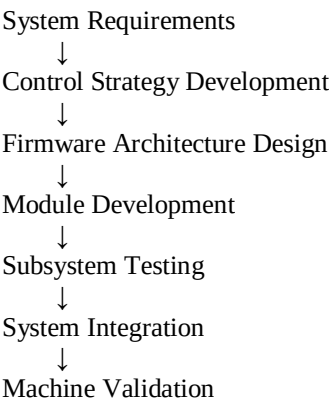
To provide the required functionality, custom embedded firmware was developed and deployed on the ESP32-S3 controller. Unlike conventional CNC systems that rely on preconfigured firmware and G-code interpretation, the proposed system utilizes application-specific control software developed using the PlatformIO development environment.

The custom software architecture was selected to provide:

- direct control of frosting operations,
- simplified implementation of frosting routines,
- flexible subsystem integration,
- easier customization,
- support for future machine enhancements.

The firmware therefore serves as the central control layer responsible for coordinating all machine operations.

Figure 3.xx: Software Development Workflow



3.16.1 Software Design Requirements

The software was required to coordinate all machine activities while maintaining synchronized frosting deposition.

The primary software requirements are summarized in Table 3.xx.

Table 3.xx Software Functional Requirements

Requirement	Purpose
Motion Coordination	Synchronize X, Y, Z and Theta axes
Extrusion Control	Control frosting delivery
Homing Operations	Establish machine reference positions
Sensor Monitoring	Detect switches and faults
Process Sequencing	Execute frosting routines
Communication	User interaction and debugging
Expandability	Future software enhancements

The software architecture was therefore developed using a modular approach to improve maintainability and simplify future upgrades.

3.16.2 Software Development Environment

Firmware development was performed using the PlatformIO development environment.

PlatformIO was selected due to:

- support for ESP32-S3 devices,
- integrated build system,
- library management capabilities,
- debugging support,
- scalability for larger embedded projects.

The development environment provides greater flexibility than preconfigured CNC firmware by allowing complete customization of machine behavior.

Table 3.xx Software Development Tools

Tool	Purpose
PlatformIO	Firmware development
ESP32-S3 Framework	Embedded execution platform
C/C++ Programming Language	Firmware implementation
Git	Version control
Serial Monitor	Debugging and testing

Figure 3.xx: Software Development Environment

Insert screenshot of:

- PlatformIO project structure,
- development environment,
- source code organization.

3.16.3 Software Architecture

The firmware was structured into independent functional modules to improve readability, maintainability, and subsystem isolation.

The major software modules include:

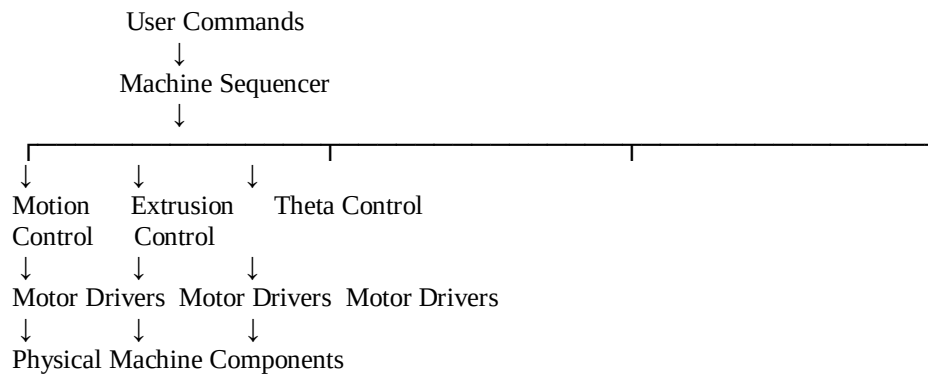
- Motion Control Module,
- Extrusion Control Module,
- Theta-Axis Control Module,
- Sensor Management Module,
- Communication Module,
- Machine Sequencing Module.

Each module performs a specific task while communicating with other modules through shared system states and control commands.

Table 3.xx Software Modules

Module	Function
Motion Control	X, Y and Z axis movement
Theta Control	Cake rotation control
Extrusion Control	Syringe actuation
Sensor Management	Input monitoring
Communication	External interaction
Machine Sequencing	Operational coordination

Figure 3.xx: Firmware Architecture Block Diagram



3.16.4 Motion Control Strategy

The frosting process requires coordinated movement of multiple machine axes.

Unlike conventional CNC machining operations, frosting quality depends on maintaining synchronization between:

- nozzle position,
- cake rotation,
- extrusion flow rate.

The software therefore implements coordinated motion scheduling routines that ensure frosting is deposited consistently throughout the frosting process.

Motion coordination includes:

Top Surface Frosting

Requires:

- X-axis motion,
- Y-axis motion,
- extrusion control.

Side-Wall Frosting

Requires:

- Z-axis motion,
- Theta-axis rotation,
- extrusion control.

Combined Operations

Requires synchronization between:

- translational movement,
- rotational movement,
- frosting flow.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from the design, fabrication, and integration of the modified automated cake frosting machine. The chapter focuses on the final machine configuration, fabricated assemblies, subsystem specifications, engineering calculations, and system-level outcomes obtained from the design process.

The machine was developed to address limitations identified in the previous frosting system, particularly those related to frosting extrusion capability, structural rigidity, rotational synchronization, and the handling of high-viscosity buttercream frosting. The resulting machine integrates a Cartesian flying gantry motion system, a Theta-axis rotary platform, a positive displacement piston extrusion mechanism, a custom electronic control system, and an ESP32-S3 based control architecture.

The chapter presents the final machine realization, subsystem results, engineering calculations, and discussion of the implications of the obtained design outcomes.

4.2 Cake Frosting Machine

4.2.1 CAD Model

Figure 4.1: CAD Isometric View

The completed design provides a Cartesian flying gantry architecture with integrated rotational cake positioning and piston-driven frosting extrusion.

4.2.2 Fabricated Machine

Following completion of the design process, the machine was fabricated and assembled into a functional prototype.

Figure 4.2: Fabricated Machine – Front View

Figure 4.3: Fabricated Machine – Isometric View

The fabricated machine closely follows the developed CAD model while incorporating practical manufacturing considerations associated with material availability, fabrication methods, and subsystem integration.

4.2.3 Major Machine Components

Table 4.1: Major Machine Components

Component	Function
Structural Frame	Supports all machine subsystems
X-Axis Assembly	Nozzle positioning
Y-Axis Assembly	Gantry translation
Z-Axis Assembly	Vertical cake positioning
Theta Axis	Cake rotation
Extrusion System	Frosting delivery
ESP32-S3 Controller	Machine control
PCB Assembly	Electrical integration
Power System	Power distribution

Discussion

The completed machine successfully integrates multiple mechanical, electrical, and software subsystems into a single automated frosting platform. The modular architecture simplifies maintenance and future modifications while providing sufficient workspace for frosting cakes up to the target dimensions specified during the design phase.

4.3 Structural Frame Results

4.3.1 Structural Frame Design

The machine frame was designed to provide a rigid support structure for the motion system, extrusion assembly, and rotary cake platform.

Figure 4.4: Structural Frame CAD Model

Figure 4.5: Fabricated Structural Frame

4.3.2 Structural Specifications

Table 4.2: Frame Specifications

Parameter	Value
Material	Mild Steel
Length	450 mm
Width	450 mm
Height	400 mm
Fabrication Method Welded Construction	

4.3.3 Discussion

The welded mild steel frame provided a rigid foundation for the machine and successfully supported the gantry system, extrusion mechanism, and rotary platform. The use of welded construction reduced structural flexibility and improved resistance to vibration during operation. The cubic frame configuration also simplified component mounting and subsystem alignment.

4.4 Motion System Results

4.4.1 Motion System Overview

The motion system consists of X-axis, Y-axis, and Z-axis positioning mechanisms integrated into a Cartesian flying gantry arrangement.

Figure 4.6: Motion System CAD Assembly

Figure 4.7: Fabricated Motion System

4.4.2 Motion System Specifications

Table 4.3: Motion System Configuration

Axis	Drive Method	Motor Type
X-Axis	Belt Drive	NEMA 17
Y-Axis	Belt Drive	NEMA 17
Z-Axis	Dual Lead Screw 2 × NEMA 17	
Theta-Axis		

4.4.3 Motion Resolution Analysis

Calculation 4.1: X-Axis Resolution

- motor step angle
- microstepping value
- pulley tooth count
- belt pitch

Calculate:

Steps/mm

Calculation 4.2: Y-Axis Resolution

Calculation 4.3: Z-Axis Resolution

- lead screw pitch
- motor step angle
- microstepping

Calculate:

Steps/mm

4.4.4 Discussion

The Cartesian architecture provides direct positional control and simplifies motion planning. The use of lead screw actuation on the Z-axis improves positional stability while belt-driven X and Y axes reduce moving mass and support faster positioning.

4.5 Theta-Axis Rotary Platform Results

4.5.1 Theta-Axis Assembly

Figure 4.8: Theta-Axis CAD Model

Figure 4.9: Fabricated Theta-Axis Assembly

4.5.2 Rotary Platform Specifications

Table 4.4: Theta-Axis Specifications

Parameter	Value
Platform Diameter	
Drive Method	
Motor	
Bearing Arrangement	

4.5.3 Rotary Motion Analysis

Calculation 4.4: Required Angular Velocity

Calculation 4.5: Rotary Torque Requirement

4.5.4 Discussion

The Theta-axis enables continuous frosting deposition around cylindrical cake surfaces. Coordination between rotational motion and nozzle translation is essential to maintain consistent frosting coverage and prevent overlap or surface gaps.

4.6 Extrusion System Results

4.6.1 Extrusion System Overview

The extrusion subsystem was developed as a positive displacement piston mechanism designed to handle high-viscosity buttercream frosting.

Figure 4.10: Extrusion System CAD Assembly

Figure 4.11: Fabricated Extrusion System

Figure 4.12: Extrusion System Exploded View

4.6.2 Extrusion System Specifications

Table 4.5: Extrusion System Parameters

Parameter	Value
Reservoir Diameter	106 mm
Reservoir Material	PPR
Operation Mode	Bottom-Up Extrusion
Lead Screws	2
Plunger Thickness	20 mm

4.6.3 Reservoir Analysis

Calculation 4.6: Reservoir Cross-Sectional Area

Insert full calculation using:

$$A = \pi D^2/4$$

Calculation 4.7: Reservoir Volume

- reservoir height
- reservoir diameter

storage capacity.

4.6.4 Extrusion Force Analysis

Calculation 4.8: Required Extrusion Pressure

- literature values
- nozzle parameters
- tubing parameters

Calculation 4.9: Required Piston Force

$$F = P \times A$$

This calculation demonstrates the force required to extrude buttercream through the delivery system.

4.6.5 Lead Screw Torque Analysis

Calculation 4.10: Lead Screw Torque Requirement

- lead screw pitch
- efficiency
- force requirement

motor torque.

Calculation 4.11: NEMA 17 Capability Assessment

- available motor force
- required extrusion force

suitability?

4.6.6 Discussion

The results indicate that buttercream frosting requires significantly higher extrusion forces than low-viscosity frosting materials. The large reservoir diameter improves frosting storage capacity but substantially increases piston force requirements. Consequently, the extrusion mechanism represents one of the most mechanically demanding subsystems within the machine.

4.7 Electronics and Control System Results

4.7.1 PCB Development

Figure 4.13: PCB Schematic

Figure 4.14: PCB Layout

Figure 4.15: Fabricated PCB

Table 4.6: PCB Functional Features

Feature	Function
ESP32-S3 Interface	Controller Hosting
Motor Driver Connections	Motion Control
Sensor Interfaces	Monitoring
Power Distribution	Voltage Routing

4.7.2 Discussion

The custom PCB reduced wiring complexity and improved subsystem integration by consolidating electrical connections into a centralized interface.

4.8 Embedded Software Results

4.8.1 Software Architecture

Figure 4.16: Firmware Architecture

Table 4.7: Software Modules

Module	Function
Motion Control	Axis positioning
Extrusion Control	Frosting delivery
Theta Control	Cake rotation
Communication	User interaction
Sensor Management	Monitoring

4.8.2 Discussion

The use of custom firmware developed in PlatformIO provides greater flexibility than traditional CNC firmware by allowing frosting-specific motion routines, direct subsystem integration, and customized machine behaviour without reliance on G-code interpretation.

4.9 System Integration Results

Figure 4.17: Fully Integrated Machine

Figure 4.18: Fully Integrated CAD Assembly

Table 4.8: Integrated Subsystems

Subsystem	Status
Structural Frame	Complete
Motion System	Complete
Extrusion System	In Progress
Electronics	Complete
Firmware	In Progress
Calibration	Pending

4.9.1 Discussion

System integration successfully combined the mechanical, electrical, and software subsystems into a unified frosting platform. Integration activities highlighted the importance of subsystem alignment, wiring management, and motion synchronization in achieving reliable machine operation.

4.10 Design Outcomes and Improvements

Table 4.9: Comparison Between Previous and Modified System

Feature	Previous System	Modified System
Frosting Material	Whipped Cream	Buttercream
Extrusion Method	Gravity-Based	Positive Displacement
Rotary Motion	Limited	Theta-Axis
Structural Frame	Welded Structure	Welded Steel Frame
Controller	Arduino	ESP32-S3

The modified machine addresses key limitations identified in the earlier design by incorporating improved extrusion capability, enhanced structural rigidity, synchronized rotational motion, and a dedicated control architecture capable of supporting future machine development.

4.11 Summary

This chapter presented the final machine realization, subsystem specifications, engineering calculations, and integration outcomes obtained from the design and fabrication of the automated cake frosting machine. The results demonstrate the feasibility of combining Cartesian motion control, rotational cake positioning, and positive displacement extrusion to support automated buttercream frosting operations. The following chapter presents system testing, calibration, and performance evaluation.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

The aim of this project was to design and develop a modified automated cake frosting machine capable of performing controlled frosting deposition on cylindrical cake geometries using buttercream frosting. The project addressed limitations identified in the previous frosting system, particularly those associated with frosting extrusion capability, rotational frosting synchronization, structural rigidity, and controlled volumetric deposition.

A moving Cartesian gantry architecture was successfully adopted as the primary motion platform due to its simplified kinematic structure, positional accuracy, and compatibility with multi-axis motion control. The inclusion of a dedicated Theta-axis rotary platform enabled synchronized rotational frosting application on cylindrical cake side walls, improving frosting continuity and surface coverage.

To address the high viscosity and non-Newtonian characteristics of buttercream frosting, a piston-driven positive displacement extrusion mechanism was developed. The selected extrusion approach provided a practical solution for generating the pressures required for controlled frosting delivery while maintaining consistent volumetric displacement.

The machine structure was designed to provide adequate rigidity for frosting operations while supporting integration of the motion system, extrusion subsystem, rotary platform, and electrical components. The modular design approach additionally improved manufacturability, subsystem accessibility, and future modification capability.

A custom embedded motion-control system based on the ESP32-S3 platform was developed using the PlatformIO development environment. The controller architecture enabled coordinated operation of the motion axes, extrusion mechanism, and rotary platform through application-specific control routines rather than conventional CNC G-code-based operation. This approach provided greater flexibility for implementing frosting-specific machine behaviors and synchronization strategies.

The completed machine demonstrates the feasibility of integrating mechatronic systems, positive-displacement food extrusion, embedded motion control, and synchronized multi-axis operation within a single automated frosting platform. The developed system provides a foundation for further research and development in automated culinary manufacturing and robotic food-deposition applications.

5.3 Recommendations

Although the developed frosting machine achieved the intended design objectives, several improvements may further enhance machine performance, reliability, and automation capability.

The following recommendations are proposed for future work:

1. Experimental rheological characterization of buttercream frosting should be conducted to obtain accurate viscosity, yield stress, and flow-property data for improved extrusion modelling and pressure prediction.
2. Closed-loop feedback systems incorporating rotary encoders, load sensors, or linear position sensors should be integrated to improve motion accuracy, synchronization, and extrusion consistency.
3. Additional investigations should be conducted to evaluate the influence of frosting temperature, nozzle diameter, extrusion speed, and rotational velocity on frosting surface quality and deposition stability.
4. Automated frosting-thickness measurement methods may be incorporated to improve calibration procedures and frosting repeatability.
5. Food-contact components should be redesigned using food-grade stainless steel materials to improve sanitation, durability, and suitability for commercial food-processing environments.
6. Future versions of the machine may incorporate machine-vision systems for cake detection, automatic alignment, and frosting quality assessment.
7. Adaptive extrusion control algorithms may be developed to automatically compensate for variations in frosting viscosity and flow resistance during operation.
8. Wireless monitoring and remote-control capabilities may be expanded through web-based user interfaces and mobile-device integration.
9. Further testing should be conducted using different cake sizes, frosting formulations, and decorative patterns to evaluate system versatility and long-term operational reliability.
10. Future development may investigate decorative pattern generation and programmable frosting path planning to expand the machine's functionality beyond uniform surface frosting applications.